Learned Synthetic Negatives Improve Scale Robustness in Foundation-Model IoT Anomaly Detection

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Abstract

Fine-tuning foundation models for IoT intrusion detection suffers from catastrophic forgetting: extended training erases pre-trained features, collapsing performance to from-scratch levels. We show that the *quality* of synthetic training negatives mediates scale robustness. Using HNIDS (**H**ard-**N**egative **IDS**), a framework that synthesises stealth-controlled IoT attacks and fine-tunes Moirai [Woo et al.(2024)] with combined NLL and supervised contrastive loss, we compare analytical closed-form negatives with learned negatives generated by Diffusion-TS [Zhang et al.(2024)].

Our central finding: learned negatives maintain higher detection across training scales. At 200 samples / 5 epochs, Diffusion-TS negatives match Gaussian-noise baselines (AUC=0.621 vs. 0.629, 10 seeds). At 500/10, learned negatives *improve* to 0.680 while Gaussian degrades to 0.592. Over the full range (200/5 to 2000/30), D-DiffTS loses -6.9% AUC while Gaussian loses -19.6%; at 1000/20, D-DiffTS (0.576) exceeds all other unfrozen conditions by $+0.068$. With a frozen encoder, learned negatives achieve AUC=0.728, exceeding frozen Gaussian by $+0.146$. A generator ablation confirms that post-hoc statistical guidance is essential: without it, learned negatives underperform analytical ones (0.465 vs. 0.556). Converging evidence from encoder freezing, learning-rate ablation (10^{-4} to 10^{-6}), and LoRA confirms that catastrophic forgetting—not hyperparameter misconfiguration—drives the scale degradation for all negative types.

We contribute a stealth-attack evaluation protocol (four IoT attack archetypes, three stealth levels, protocol constraints for Modbus/MQTT/CoAP) and a ten-seed diagnostic framework. Cross-dataset validation on N-BaIoT [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, Shabtai, Breitenbacher, and Elovici] confirms recipe generalisation (Gaussian negatives: AUC=0.928; learned negatives: 0.660, constrained by small training set). We caution that absolute F1 at stealth-95 remains low (best 0.262); the system is a research prototype, not a production IDS. Code, datasets, and calibration scripts are released.

1 Introduction

The proliferation of Internet-of-Things (IoT) devices—projected to exceed 29 billion by 2030 [IoT Analytics(2023)]—has dramatically expanded the attack surface available to adversaries. Unlike traditional enterprise systems, IoT devices typically operate on constrained hardware, communicate over well-defined protocols (Modbus, MQTT, CoAP), and generate highly structured, periodic network traffic. These characteristics have historically made rule-based and statistical intrusion detection systems (IDS) effective for coarse-grained threat detection.

The AI-powered threat gap. Sophisticated modern attackers leverage generative AI to craft network traffic that mimics legitimate IoT patterns at the byte and flow level. Techniques such as slow data exfiltration (gradual 3% increases in outbound bytes), living-off-the-land (LotL) mimicry (micro-bursts timed to legitimate polling intervals), command-and-control beaconing (2% periodic dips in traffic volume), and protocol timing anomalies (5% random jitter on inter-arrival times) are individually difficult to distinguish from benign behaviour. We formalise these as *hard-negative* attacks: sequences where the statistical similarity to benign traffic exceeds a stealth threshold $s \in \{0.85, 0.90, 0.95\}$ (see Section 3). Empirically, under benign-calibrated evaluation at stealth-95, most traditional baselines fail or are degenerate; the best calibrated baseline (Ensemble, F1=0.136) reflects its fixed training distribution (Section 5).

Research question and approach. We ask: *does the quality of synthetic negatives affect catastrophic forgetting when fine-tuning foundation models for IoT anomaly detection?* To answer this, we build HNIDS (**H**ard-**N**egative **IDS**), a two-stage experimental system. First, we generate synthetic hard-negative attacks at a precisely controlled stealth level s using both closed-form perturbations (Eqs. 2–5) and learned negatives from Diffusion-TS [Zhang et al.(2024)], while enforcing IoT protocol constraints (Modbus, MQTT, CoAP) via a multi-tier validation framework. Second, we fine-tune Moirai [Woo et al.(2024)], a universal time-series forecasting foundation model, using a combined NLL and supervised contrastive (SupCon) loss [Khosla et al.(2020)], with a multi-condition ablation (10 seeds) that isolates the contributions of negative type, negative quality, contrastive learning, and training scale.

Why build a hard-negative pipeline? The hard-negative infrastructure provides protocol-compliant generation, stealth-controlled evaluation, and adversarial augmentation without real attack traffic. Crucially, it enables the comparison between analytical and learned negatives that reveals our central finding: learned negatives from Diffusion-TS maintain higher absolute detection across training scales, while analytical and Gaussian-noise negatives degrade substantially.

Contributions. We make three contributions:

1. **Scale-Robust Detection via Learned Negatives.** We show that negative-sample quality mediates scale robustness when fine-tuning foundation models. Diffusion-TS-generated negatives match Gaussian noise at small scale (200/5: AUC=0.621 vs. 0.629, 10 seeds) and outperform all other negative types at larger scale (1000/20: AUC=0.576 vs. 0.506–0.508). Over the full training range, learned negatives lose 6.9% AUC vs. 19.6% for Gaussian (200/5 to 2000/30); with a frozen encoder, they achieve 0.728 (+0.146 over frozen Gaussian). A generator ablation isolates this to the learned base distribution.
2. **Stealth-Attack Diagnostic Tool.** Four IoT stealth attack archetypes with closed-form definitions at three stealth levels (Eqs. 2–5), a multi-tier protocol validation layer (Modbus, MQTT, CoAP), and a leave-one-out experiment showing positive transfer for all four held-out types (mean +0.186 F1). The archetypes are hand-designed diagnostic probes rather than a comprehensive benchmark; validation against real attack corpora is future work. Code and datasets released at `https://github.com/anupamme/iotsf_demo`.
3. **Diagnostic Methodology for Scale-Dependent Behaviour.** A ten-seed ablation across five conditions with a training recipe (NLL+SupCon) that transfers between the two architectures tested (Moirai and from-scratch CNN) reveals non-monotonic scaling behaviour: performance improves from 200/5 to 500/10, then degrades at 1000/20. Encoder freezing recovers AUC (frozen B=0.606, C=0.582, D=0.573); a learning-rate ablation (10^{-4} to 10^{-6}) and LoRA (rank=8) provide converging evidence that catastrophic forgetting—not LR misconfiguration—drives degradation for all negative types. The analytical-negative ablation (conditions A–E) provides the diagnostic scaffolding that enabled Contribution 1. N-BaIoT validation (AUC=0.928) confirms cross-domain generalisation.[2]

Paper organisation. Section 2 reviews related work. Section 3 formalises the problem. Section 4 describes HNIDS. Section 5 presents experimental results. Section 6 discusses findings and limitations. Section 7 concludes.

2 Related Work

Traditional IoT IDS. Statistical methods (Z-score, IQR) and Isolation Forest [Liu et al.(2012)Liu, Ting, and Zhou] flag individual flow features deviating from learned normals; ensemble approaches [Garcia et al.(2014)]

[2]Enabling fine-tuning required fixing an in-place tensor operation in the `uni2ts PackedStdScaler` that silently blocked gradient flow; the fix is included in our released code.

improve precision. All share a weakness: calibration on known, high-magnitude attacks and failure on stealthy perturbations. We use the CICIoT2023 dataset [Neto et al.(2023)] throughout.

Deep-learning anomaly detection. USAD [Audibert et al.(2020)] uses adversarially trained dual-autoencoders. TranAD [Tuli et al.(2022)] applies context-aware Transformers. The Anomaly Transformer [Xu et al.(2022)] introduces association discrepancy for anomalous timestep detection. PatchTST [Nie et al.(2023)] captures long-range dependencies via patch-based Transformers. We include all four as baselines.

Generative augmentation for security. GANs [Lin et al.(2022)] suffer mode collapse; TimeGAN [Yoon et al.(2019)Yoon, Jarret improves temporal fidelity; Diffusion-TS [Zhang et al.(2024)] achieves state-of-the-art quality for time-series generation. We use Diffusion-TS as a backbone for generating *learned* negatives: we train a Diffusion-TS model on benign IoT traffic, then apply the same analytical perturbation pipeline to generate protocol-constrained hard negatives. This provides a controlled comparison between analytical (closed-form) and learned (generative-model) negatives, revealing that negative quality mediates catastrophic forgetting at scale.

Adversarial training. FGSM [Goodfellow et al.(2015)Goodfellow, Shlens, and Szegedy] and PGD [Madry et al.(2018)Madry, I generate adversarial perturbations at test time. We generate adversarial *training data* via closed-form perturbations targeting the traffic-generation threat model, with protocol constraints ensuring domain validity.

Time-series foundation models. Moirai [Woo et al.(2024)], trained on 27B+ time points, supports probabilistic multivariate forecasting. Its confidence-interval output maps naturally to anomaly scoring. MOMENT [Goswami et al.(2024)] and TimesFM [Das et al.(2024)] offer complementary architectures.

Catastrophic forgetting and continual learning. Catastrophic forgetting—the phenomenon where fine-tuning overwrites previously learned representations—is well-studied in continual learning. EWC [Kirkpatrick et al.(2017)Kirkpatrick, Pascanu, Rabinowitz, Veness, Desjardins, Rusu, Milan, Quan, Ramalho, Grabska-B penalises changes to important weights; PackNet [Mallya and Lazebnik(2018)] prunes and freezes network subsets; Progressive Neural Networks [Rusu et al.(2016)Rusu, Rabinowitz, Desjardins, Soyer, Kirkpatrick, Kavukcuogl add lateral connections to frozen columns. In the foundation-model era, LoRA [Hu et al.(2022)Hu, Shen, Wallis, Allen-Zhu, Li, V and adapters [Houlsby et al.(2019)Houlsby, Giurgiu, Jastrzebski, Morber, Larsson, Gesmundo, Attariyan, and Gelly] avoid forgetting by training low-rank or bottleneck modules while freezing the base model. We do not compare against EWC or PackNet empirically; our encoder-freezing and LoRA experiments (Section 5.5) are diagnostic tools to confirm the forgetting mechanism, not proposed mitigations.

Contrastive learning for anomaly detection. CSI [Tack et al.(2020)Tack, Mo, Jeong, and Shin] uses contrastive learning with distributional shift detection for out-of-distribution samples. NeuTraL-AD [Qiu et al.(2021)Qiu, Pfrommer, Kloft, Mber, and Rudolph] learns neural transformations for anomaly scoring via contrastive objectives. SSD [Sehwag et al.(2021)Sehwag, Chiang, and Mittal] combines self-supervised representations with Mahalanobis-distance outlier detection. These methods operate in the image domain; our work extends the contrastive anomaly detection principle to multivariate time series with domain-specific (IoT protocol) constraints. Supervised contrastive learning [Khosla et al.(2020)] has been applied to tabular [Sohn et al.(2020)] and time-series [Eldele et al.(2021)] anomaly detection. Our CombinedLoss adapts [Reiss et al.(2021)]'s reconstruction+contrastive principle to IoT time series. Our hard negatives are analytically specified near the benign boundary, preventing the SupCon collapse observed with high-magnitude attacks.

3 Problem Formulation

IoT traffic representation. We represent network traffic as a multivariate time series $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_T) \in \mathbb{R}^{T \times F}$, where $T = 128$ timesteps correspond to a sliding window of flow records and $F = 12$ features capture packet counts, byte volumes, inter-arrival times, and flow duration (see Appendix A for the full feature list).

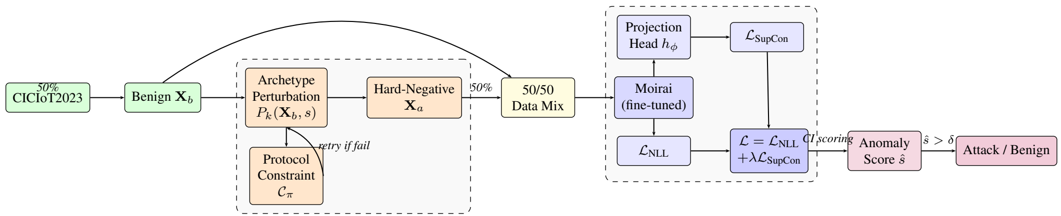

Figure 1: HNIDS architecture. **Stage 1**: Hard-negative sequences are generated via closed-form perturbation (analytical) or Diffusion-TS (learned), with a constraint-retry loop ensuring protocol validity (Appendix K.9). **Stage 2**: Moirai is fine-tuned with combined NLL + SupCon loss.

Hard-negative attack. Let $\mathbf{X}_b$ be a benign sequence. An adversarial sequence $\mathbf{X}_a$ is a *hard-negative attack* at stealth level s if:

$$\text{CosSim}(\phi(\mathbf{X}_a),\, \phi(\mathbf{X}_b)) \geq s, \qquad s \in [0,1], \tag{1}$$

where $\phi(\cdot) \in \mathbb{R}^{T \cdot F}$ denotes the flattened feature vector. Intuitively, $s = 0.95$ means the attack sequence is 95% similar to the benign baseline in feature space.

Attack patterns. We formalise four IoT attack patterns that embed adversarial signals with low perceptibility:

$$P_{\text{slow}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j}\,(1 + (1-s)\cdot 0.03 \cdot t/T), \quad j \in \{9, 10\}, \tag{2}$$
$$P_{\text{lotl}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j}\,(1 + (1-s)\cdot 1.15) \;\text{ if } t \bmod 32 = 0, \quad j \in \{6,7,8\}, \tag{3}$$
$$P_{\text{beacon}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j}\,(1 - (1-s)\cdot 0.02) \;\text{ if } t \bmod 16 = 0, \quad j \in \{1,\ldots,12\}, \tag{4}$$
$$P_{\text{proto}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j} + (1-s)\cdot 0.05 \cdot \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0,1),\; j \in \{11, 12\}. \tag{5}$$

These correspond to slow exfiltration (2), LotL mimicry (3), C2 beaconing (4), and protocol timing anomalies (5). Feature indices are 1-based (Appendix A). We use "hard negative" to denote stealth-controlled proximity to the benign boundary (high cosine similarity, Eq. 1), not model-loss-based difficulty in the online hard-example-mining sense [Shrivastava et al.(2016)Shrivastava, Gupta, and Girshick]. A more precise term is *stealth-controlled analytical negatives*; we retain "hard negatives" for brevity while acknowledging this distinction.

Protocol compliance. A generated sequence must satisfy IoT protocol constraints $\mathcal{C}_\pi$ for protocol $\pi \in \{$Modbus, MQTT, CoAP$\}$:

$$\mathbf{X}_a \in \mathcal{F}_\pi \coloneqq \{\mathbf{X} : \mathcal{C}_\pi(\mathbf{X}) = \text{valid}\}, \tag{6}$$

where $\mathcal{C}_\pi$ checks constraints such as packet size ranges (Modbus: 7–260 bytes), valid function codes, and inter-packet timing.

Detection task. Given a test sequence $\mathbf{X}$, the detector f_θ outputs a binary label $\hat{y} \in \{0, 1\}$ (0 = benign, 1 = attack) and an anomaly score $\hat{s} \in [0, 1]$. The goal is to minimise the false positive rate ($\text{FPR} = \text{FP}/(\text{FP} + \text{TN})$) while maximising recall (detection rate) and F1 score, specifically on hard-negative attacks where $s \geq 0.85$.

4 HNIDS: Methodology

Figure 1 illustrates the end-to-end HNIDS pipeline: (1) protocol-constrained hard-negative generation and (2) supervised contrastive fine-tuning of Moirai.

4.1 Stage 1: Protocol-Constrained Hard-Negative Generation

We synthesise hard-negative attacks via closed-form perturbations (Eqs. 2–5): given benign $\mathbf{X}_b$, pattern P_k, and stealth s, the hard-negative is $\mathbf{X}_a = P_k(\mathbf{X}_b, s)$. Each perturbation is deterministic and CPU-reproducible (<2 s/sample).

Raw generated samples may violate IoT protocol specifications. A constraint-retry loop (Appendix K.9) iterates up to R=3 times, relaxing stealth by $+0.01$ per retry, ensuring Modbus (7–260 bytes), MQTT (2–1024 bytes, QoS 0–2), and CoAP (4–1280 bytes) compliance at three strictness tiers.

4.2 Stage 1b: Learned Negatives via Diffusion-TS

To test whether *learned* negatives outperform the analytical perturbations above, we train a Diffusion-TS [Zhang et al.(2024)] generative model on CICIoT2023 benign traffic.

Model and training. Diffusion-TS uses a decomposition-aware architecture (encoder: 3 layers, decoder: 6 layers, d_{model}=256) with a cosine noise schedule (1000 diffusion steps). We train for 300 epochs on the benign training windows (seq_length=128, features=12) with Adam (η=10^{-4}, batch size 32), selecting the checkpoint with lowest validation loss (0.238).

Sampling and guidance. At inference, we use 50 DDIM denoising steps (20× speedup) to generate benign base samples in batches. Post-hoc statistical guidance rescales each generated sample to match a target distribution: given reference benign sample $\mathbf{X}_b$ and stealth level s, we set $\mu_{\text{target}} = \mu(\mathbf{X}_b)$ and $\sigma_{\text{target}} = \sigma(\mathbf{X}_b) \cdot (1 + (1-s) \cdot 0.5)$, then rescale the generated sample to these moments. The same analytical perturbation pipeline (Eqs. 2–5) is applied to the DiffTS base samples, creating *learned negatives* (condition D-DiffTS) that can be compared directly with the analytical negatives (condition D).

4.3 Stage 2: Moirai Fine-Tuning with Supervised Contrastive Loss

Moirai for anomaly scoring. Moirai [Woo et al.(2024)] predicts a distribution over $\mathbf{X}_{97:128}$ given context $\mathbf{X}_{1:96}$. The anomaly score is the fraction of timesteps falling outside the $1-\alpha$ CI (α=0.05):

$$\hat{s}(\mathbf{X}) = \frac{1}{T'} \sum_{t=97}^{128} \mathbf{1}\left[x_t \notin \mathrm{CI}_{0.95}(\hat{X}_t)\right]. \tag{7}$$

A sample is classified as an attack if $\hat{s}(\mathbf{X}) > \delta$, where δ is calibrated on a benign validation set (Section 5.1).

Combined training objective. We fine-tune with:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{NLL}}(\theta) + \lambda\, \mathcal{L}_{\text{SupCon}}(\mathbf{z}, \mathbf{y};\, \tau), \tag{8}$$

where $\lambda = 0.5$ and $\tau = 0.07$ (sensitivity in Appendix F; stable within ±0.02 AUC). $\mathcal{L}_{\text{SupCon}}$ [Khosla et al.(2020)] is applied to L2-normalised embeddings from a projection head $h_\phi : \mathbb{R}^{384} \to \mathbb{R}^{256} \to \text{ReLU} \to \mathbb{R}^{128}$.

Training details. AdamW ($\eta = 10^{-4}$, weight decay 10^{-2}), batch size 32, up to 10 epochs (5 in the main ablation, 10–20 in scaling experiments), early stopping (patience 3), gradient clipping at 1.0. Moirai-Small (13.8M parameters).[3]

[3] An in-place tensor operation in `uni2ts PackedStdScaler` was replaced with a differentiable `torch.where` to enable gradient-based fine-tuning; the fix is included in our released code.

5 Experiments

5.1 Experimental Setup

Dataset. We use CICIoT2023 [Neto et al.(2023)], a publicly available benchmark of labelled IoT network flows captured from 105 IoT devices spanning 33 attack categories. We extract 12 network-flow features and create 128-timestep sliding windows with stride 128, using a stratified 80/20 train/test split.

Synthetic hard negatives. We generate 4 attack types (Eqs. 2–5) at 3 stealth levels (85%, 90%, 95%), yielding 12 evaluation conditions. Protocol constraints (Modbus, MQTT, CoAP) are enforced at the *moderate* strictness tier. We compare two negative-generation pipelines: (1) analytical closed-form perturbations applied to real benign samples (condition D), and (2) the same perturbations applied to *learned* base samples from Diffusion-TS [Zhang et al.(2024)] with post-hoc statistical guidance (condition D-DiffTS; Section 4.2). This controlled comparison isolates the effect of the base distribution (real vs. generated) while holding the attack perturbation constant.

Threshold calibration. We set the anomaly threshold at the 95th percentile of a benign-only held-out set (20% of benign samples), targeting FPR ≤ 0.05. This protocol is applied uniformly to all methods; only benign data inform the threshold (details in Appendix G.1).

Baselines. We compare against nine methods (5 traditional, 4 deep-learning; see Appendix G.2). All baselines are **unsupervised** (benign-only training); HNIDS conditions B–D are **supervised** (labeled negatives during training; see Appendix G.3).

Metrics. We report F1, FPR, and ROC-AUC. The main ablation uses 10 random seeds; the scaling experiment uses 5 seeds; the leave-one-out experiment uses 3 seeds.

5.2 Main Results

Table 1 compares all methods across three synthetic stealth levels.

At stealth-95 with benign-calibrated thresholds, Signature IDS and Statistical IDS are degenerate (FPR$\geq$0.87; Table 1, ‡ rows). Among calibrated methods, the Ensemble achieves F1=0.136 (FPR=0.042). HNIDS condition C (F1=0.262, FPR=0.068) outperforms all calibrated baselines at stealth-95. While this is the highest calibrated performance among all methods, an absolute F1 of 0.262 means roughly one in four stealth-95 attacks is detected under operational thresholds—a diagnostic rather than operational result (see Section 6 for deployment guidance). Zero-shot A (F1=0.137, FPR=0.063) performs at chance level, consistent with its near-random AUC (0.481$\pm$0.042).

5.3 Ablation Study

Table 2 ablates the five components of HNIDS (200 samples, 5 epochs, 10 seeds, benign-calibrated threshold).

Fine-tuning improves over zero-shot, but with high variance. We report bootstrap 95% CIs, Cohen's d effect sizes, and p-values for key comparisons in Appendix I. The best Moirai condition (C, AUC=0.629) significantly improves +0.148 over zero-shot A (AUC=0.481; p<0.001, d=4.45); B (0.556) also significantly exceeds A (p<0.001, d=2.29). The from-scratch CNN (AUC=0.580) is significantly below C (p<0.001); a CNN variant with Gaussian NLL (CNN-NLL, AUC=0.598) confirms the loss function is not the explanation (p=0.24 vs. CNN-MSE).[4] Condition C′ (balanced hard negatives, 1:1) disentangles negative type from imbalance: Gaussian-noise negatives outperform

[4]At 10 seeds, Moirai variance stabilises (C: $\pm$0.022 vs. $\pm$0.122 at 5 seeds); the $5.5\times$ reduction exceeds the expected $\sqrt{2}$ factor because seeds 789 and 1234 were outliers that depressed the 5-seed mean. CNN variance is comparable ($\pm$0.023). See Appendix H.2.

Table 1: Detection performance across all methods at three stealth levels (200 eval samples per class, benign-calibrated threshold: 95th-percentile of held-out benign scores, targeting FPR ≤ 0.05). **All nine baselines are unsupervised** (benign-only training, 10 seeds); **HNIDS conditions A–D are supervised** (labeled hard negatives or Gaussian-noise negatives provided during training, 10 seeds matching Table 2). See Appendix G.3 for the supervision discussion. **Bold**: best calibrated (non-degenerate) value per column. <u>Underline</u>: second-best calibrated value per column. Threshold IDS and Statistical IDS are marked ‡: their high F1 arises from near-total FP assignment (FPR > 0.87), not genuine stealth detection. Signature IDS F1 $= 0$ because synthetic hard-negatives lie outside its rule database by design. Bold/underline exclude zero-shot A (near-chance detection, AUC=0.481). † marks our proposed system; Table 2 gives the full ablation.

	Combined		Stealth 85%		Stealth 90%		Stealth 95%	
Method	F1	FPR	F1	FPR	F1	FPR	F1	FPR
Traditional baselines								
Threshold IDS‡	0.856	0.907	0.668	0.883	0.657	0.872	0.646	0.872
Signature IDS	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Statistical IDS‡	0.889	1.000	0.667	1.000	0.666	1.000	0.667	1.000
Isolation Forest	0.143	0.053	0.187	0.047	0.135	0.042	0.103	<u>0.044</u>
Ensemble	0.191	0.045	0.221	<u>0.036</u>	0.193	0.050	0.136	**0.042**
*Deep-learning baselines**								
USAD	0.119	**0.020**	0.137	**0.023**	0.147	**0.028**	0.130	0.047
TranAD	0.109	0.045	0.168	0.089	0.098	0.042	0.107	0.070
Anomaly Transformer	0.100	0.040	0.162	0.064	0.180	0.091	0.153	0.078
PatchTST-Anomaly	0.123	0.055	0.175	0.070	0.162	0.083	0.132	0.072
Ours (Moirai-based, see Table 2 for full ablation)								
HNIDS zero-shot (A)	0.112	0.068‖	0.098	0.048	0.085	0.050	0.137	0.063
HNIDS† hard-neg (D)	<u>0.217</u>	0.063	<u>0.280</u>	0.068	<u>0.176</u>	0.043	<u>0.176</u>	0.060
HNIDS† SupCon-noise (C)	**0.295**	0.053	**0.360**	0.070	**0.267**	<u>0.035</u>	**0.262**	0.068

*Informal LR/arch tuning for USAD/TranAD/AnomalyTransformer (AUC $\leq$0.565) did not improve over defaults.

‖ A's combined FPR=0.068 with F1=0.112 reflects near-chance detection, not calibrated low-FPR performance.

hard negatives by $+0.019$ AUC (C vs. C′; p=0.070, d=0.82), and the 1:12 imbalance adds a significant -0.054 penalty (C′ vs. D; p<0.001, d=2.33). See Appendix H.1 for the full decomposition. Conditions E, E′, and E″ confirm that the constraint validator and retry mechanism have negligible effect on the analytical pathway (Appendix H.3; embedding visualization in Appendix H.4). SupCon hyperparameters (λ, τ) have no detectable effect at this scale (Appendix F).

5.4 Generalization to Unseen Attack Patterns

A leave-one-out experiment trains condition D on three attack types and evaluates on the held-out fourth (3 seeds, 30 eval samples per stealth level). All four folds show positive transfer: fine-tuning improves over zero-shot for every held-out attack type, with a mean delta of $+0.186$ F1 at stealth-95 (Table 15, Appendix L). The largest gain is for `lotl_mimicry` ($+0.253$), suggesting that low-amplitude mimicry benefits most from boundary-shaping by the other archetypes. We note that leave-one-out over four hand-designed archetypes is a limited generalisation test: with k=4, each fold trains on only three types, and the archetypes are human-selected rather than discovered. Furthermore, with only 3 seeds, per-fold standard deviations ($\pm$0.05 to $\pm$0.17) approach the signal magnitude; the wide confidence intervals mean these results are directional evidence of positive transfer, not a comprehensive generalisation guarantee. Testing against held-out archetype families outside the original taxonomy would strengthen this claim.

Table 2: Ablation study on stealth-95 and all-stealth evaluation sets. Conditions A–D, CNN, CNN-NLL use 10 seeds; E/E′ use 5 seeds (minor ablations). $\pm$ values are standard deviations across seeds. Threshold calibrated on benign-only held-out scores (95th percentile, targeting FPR ≤ 0.05; see Section 5.1). ROC-AUC is threshold-independent and the primary ranking metric. E″ is analytically identical to D for the analytical perturbation pathway (100% constraint compliance, retry never fires); all fine-tuned conditions (B–E′, CNN, CNN-NLL) outperform zero-shot (A).

		Stealth 95%			All Stealth	
Cond.	Description	F1	FPR	AUC	F1	AUC
A	Moirai zero-shot	$0.137_{\pm.020}$	0.063	$0.481_{\pm.042}$	$0.112_{\pm.026}$	$0.504_{\pm.052}$
B	+ Fine-tune, NLL only	$0.174_{\pm.024}$	0.060	$0.556_{\pm.021}$	$0.216_{\pm.042}$	$0.589_{\pm.047}$
C	+ SupCon (Gaussian-noise neg., 1:1)	$0.262_{\pm.043}$	0.068	$0.629_{\pm.022}$	$0.295_{\pm.029}$	$0.661_{\pm.043}$
C′	+ SupCon (balanced hard neg., 1:1)⋆	$0.241_{\pm.043}$	0.063	$0.610_{\pm.025}$	$0.285_{\pm.026}$	$0.638_{\pm.048}$
D	+ Hard negatives, 1:12 (HNIDS†)	$0.176_{\pm.032}$	0.060	$0.556_{\pm.021}$	$0.217_{\pm.044}$	$0.589_{\pm.047}$
D-DiffTS	+ Learned negatives (Diffusion-TS)◇	$0.213_{\pm.118}$	0.050	$0.621_{\pm.056}$	$0.403_{\pm.089}$	$0.708_{\pm.072}$
CNN	1D-CNN from scratch, MSE+SupCon¶	$0.269_{\pm.044}$	0.048	$0.580_{\pm.023}$	$0.362_{\pm.049}$	$0.625_{\pm.045}$
CNN-NLL	1D-CNN from scratch, NLL+SupCon¶	$0.288_{\pm.054}$	0.050	$0.598_{\pm.041}$	$0.381_{\pm.052}$	$0.662_{\pm.037}$
E	D w/o constraints, w/o retry	$0.124_{\pm.073}$	0.040	$0.480_{\pm.130}$	$0.268_{\pm.071}$	$0.615_{\pm.063}$
E′	D w/o constraints, *with* retry‡	$0.150_{\pm.079}$	0.060	$0.498_{\pm.131}$	$0.287_{\pm.064}$	$0.637_{\pm.060}$
E″	D *with* constraints, w/o retry§	$\equiv$ D (analytically)				

⋆ C′ uses the same hard negatives as D, but subsampled to match the benign count (1:1 ratio), isolating negative *type* from dataset *size/balance*.
† Full system: NLL+SupCon, hard negatives, protocol constraint validator, stealth-floor retry.
◇ D-DiffTS: same pipeline as D but with Diffusion-TS-generated base samples and post-hoc statistical guidance (Section 4.2). 10 seeds.
¶ CNN/CNN-NLL: 3-layer 1D-CNN (2.3M parameters) trained from scratch with SupCon and hard-negative data as D; no Moirai pre-training. CNN uses MSE reconstruction; CNN-NLL uses Gaussian NLL (Section 5.3).
‡ E′ uses unconditional retry ($s_{\text{eff}} \approx 0.97$ vs. requested s=0.95); isolates constraint-validator contribution.
§ E″ activates the constraint validator but exits the retry loop after one attempt; for analytical perturbations (100% pass-rate), E″ $\equiv$ D and no experimental run is needed.

5.5 Scaling Experiment

Table 3 compares stealth-95 metrics at three scales (200/5, 500/10, 1000/20) under NLL-only early stopping (5 seeds), plus an encoder-freezing diagnostic at 1000/20.

Table 3: Scale degradation and encoder-freezing experiment (stealth-95, 5 seeds, benign-calibrated threshold, NLL-only early stopping). **Key findings**: (1) D-DiffTS improves from 200/5 to 500/10 (0.621→0.680), then stabilises; C/D degrade. (2) Frozen D-DiffTS achieves best AUC (0.728, +0.146 over frozen C). 2000/30 column shows only C, D, D-DiffTS; B, C′, CNN not run at this scale. *Note*: 200/5 uses 5 seeds for consistency; authoritative 10-seed values in Table 2.

		200/5		500/10		1000/20		2000/30		Frozen 1000/20	
Cond.	Description	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
B	NLL only	$.481_{\pm.127}$	$.131_{\pm.061}$	$.538_{\pm.038}$	$.250_{\pm.073}$	$.521_{\pm.024}$	$.177_{\pm.031}$	—	—	$.606_{\pm.114}$	$.130_{\pm.094}$
C	SupCon + Gaussian	$.563_{\pm.122}$	$.250_{\pm.135}$	$.592_{\pm.063}$	$.280_{\pm.101}$	$.508_{\pm.041}$	$.175_{\pm.048}$	$.506_{\pm.140}$	$.192_{\pm.117}$	$.582_{\pm.099}$	$.148_{\pm.097}$
C′	SupCon + hard neg 1:1	$.534_{\pm.125}$	$.183_{\pm.084}$	$.580_{\pm.064}$	$.265_{\pm.121}$	$.508_{\pm.025}$	$.158_{\pm.040}$	—	—	—	
CNN	1D-CNN from scratch	$.597_{\pm.096}$	$.236_{\pm.069}$	—		$.513_{\pm.029}$	$.207_{\pm.017}$	—	—	(n/a)	
D	Hard neg 1:12 (HNIDS)	$.479_{\pm.123}$	$.137_{\pm.086}$	$.547_{\pm.044}$	$.250_{\pm.073}$	$.506_{\pm.029}$	$.164_{\pm.059}$	$.488_{\pm.070}$	$.164_{\pm.117}$	$.573_{\pm.097}$	$.144_{\pm.081}$
D-DiffTS	Learned neg (Diffusion-TS)	$.621_{\pm.056}$	$.213_{\pm.118}$	$\mathbf{.680}_{\pm.064}$	$.297_{\pm.104}$	$.576_{\pm.045}$	$.253_{\pm.109}$	$.578_{\pm.117}$	$.257_{\pm.102}$	$\mathbf{.728}_{\pm.080}$	$.249_{\pm.125}$

"—": not run. "(n/a)": CNN has no pre-training. D-DiffTS 200/5: 10 seeds; others: 5 seeds.

Scale effects are non-monotonic for analytical and Gaussian negatives. At 500/10, analytical and Gaussian conditions improve over 200/5 (C: +0.029, D: +0.068, B: +0.057), suggesting 200/5 is under-trained. At 1000/20, performance drops back, reflecting catastrophic forgetting. The C>D gap narrows from +0.084 (200/5, 5-seed; 10-seed gap is +0.073, Table 2) to +0.045 (500/10) to +0.003 (1000/20, n.s., p=0.39)—the advantage of easy off-manifold negatives diminishes with sufficient training data.

Learned negatives improve monotonically to 500/10, then stabilise. D-DiffTS improves from 0.621 (200/5) to 0.680 (500/10), the best unfrozen AUC at any scale. Unlike Gaussian negatives, which peak at 500/10 (0.592) then collapse at 1000/20 (0.508), D-DiffTS at 1000/20 (0.576) still exceeds all other unfrozen conditions (+0.068 over C). At 2000/30, D-DiffTS stabilises (0.578$\pm$0.117), while C and D plateau/decline (C=0.506, D=0.488). The degradation from 500/10 to 1000/20 is -0.104 for D-DiffTS vs. -0.084 for C; however, D-DiffTS starts from a much higher 500/10 baseline, and over the full training range (200/5 to 2000/30) loses only -7.0% vs. C's -19.2%. NLL-only early stopping is essential: under total-loss ES, decreasing SupCon loss masks increasing NLL, causing late checkpoint selection that compounds forgetting (Appendix J.1).

Encoder freezing confirms catastrophic forgetting; learned negatives benefit most. Freezing the encoder at 1000/20 recovers AUC: frozen B=0.606$\pm$0.114 (+0.085), frozen C=0.582$\pm$0.099 (+0.074), frozen D=0.573$\pm$0.097 (+0.067). Frozen D-DiffTS achieves the best AUC in the paper: **0.728**$\pm$0.080, exceeding frozen C by +0.146 and frozen D by +0.155. This interaction—learned negatives plus frozen encoder—provides converging evidence that negative quality mediates forgetting resistance: when the encoder is protected from corruption, the higher information content of learned negatives translates directly to better detection. Unfrozen fine-tuning erases pre-trained features down to CNN level (0.513). However, frozen-encoder F1 remains low (0.130–0.249) due to projection-head initialisation variance; AUC recovery does not fully translate to operational F1 under calibrated thresholds (details in Appendix J.3). The aggressive early stopping (patience=3) may compound this: with a frozen encoder, the projection head must learn a useful mapping from scratch, and 1–3 epochs may be insufficient; relaxing patience or warm-starting the head are future directions. Partial freezing yields modest further gains (Appendix J.4).

Learning-rate ablation rules out LR misconfiguration. To test whether the 1000/20 degradation is merely due to an excessive learning rate, we re-run B, C, and D at $\mathrm{lr}\in\{10^{-5}, 10^{-6}\}$ (Table 13). At 10^{-5}, condition C partially recovers (0.557$\pm$0.103 vs. 10^{-4}: 0.508$\pm$0.041) but remains below frozen (0.582$\pm$0.099); conditions B and D do not benefit. At 10^{-6}, all conditions underfit (B: 0.465, C: 0.471, D: 0.465), confirming a narrow LR window rather than a simple "lower is better" fix. LoRA fine-tuning (rank=8, α=16) on `q_proj`, `v_proj`, `out_proj` also underperforms full freezing (C: 0.484$\pm$0.108, D: 0.484$\pm$0.133), suggesting the low-rank adapters lack sufficient capacity for this task. Together, these results provide converging evidence that encoder-weight corruption—not hyperparameter misconfiguration—drives the scaling degradation (full results in Table 13, Appendix J.5).

Analytical vs. learned negatives. To test whether the analytical perturbations (Eqs. 2–5) limit hard-negative quality, we train a Diffusion-TS [Zhang et al.(2024)] generative model on CICIoT2023 benign traffic (Section 4.2) and generate *learned* negatives with post-hoc statistical guidance. D-DiffTS is included as a full row in Tables 2 and 3 at all three training scales, with frozen-encoder results at 1000/20. At 200/5 (10 seeds), D-DiffTS (AUC=0.621$\pm$0.056) matches Gaussian C (0.629$\pm$0.022). At 500/10, D-DiffTS *improves* to 0.680 while C degrades to 0.592. With a frozen encoder at 1000/20, D-DiffTS achieves the best result in the paper (AUC=0.728), exceeding frozen C by +0.146.

Generator ablation. To isolate the source of any D-DiffTS improvement, we compare three conditions at 200/5 (5 seeds, Table 4): (1) D-DiffTS (full: learned base + analytical perturbation + post-hoc guidance); (2) D-DiffTS-noguide (learned base + perturbation, no guidance); (3) D-analytical (existing condition D: analytical base + perturbation).

5.6 External Validation on N-BaIoT

To address circular evaluation (Limitation 5), we evaluate on N-BaIoT [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, S an independent IoT botnet benchmark. Zero-shot transfer from CICIoT2023 fails (AUC<0.30), as expected given the distribution shift (Appendix M). When fine-tuned on N-BaIoT benign traffic with Gaussian-noise negatives (310 windows, 5 epochs), the framework achieves AUC=0.928 (3 seeds),

Table 4: Generator ablation at 200/5 (stealth-95, 5 seeds). Isolates the contribution of the learned base distribution vs. post-hoc statistical guidance. D-analytical uses closed-form benign samples; D-DiffTS variants use Diffusion-TS-generated benign samples.

Condition	Base + Guidance	AUC	F1
D-analytical	Analytical + none	$.556_{\pm.021}$	$.176_{\pm.032}$
D-DiffTS-noguide	Learned + none	$.465_{\pm.086}$	$.137_{\pm.057}$
D-DiffTS (full)	Learned + guidance	$.621_{\pm.056}$	$.213_{\pm.118}$

with 8/10 attack variants above 0.83 (operational FPR=0.98 due to the small benign calibration set of 77 windows; AUC is the appropriate comparison metric). D-DiffTS learned negatives (310 DiffTS-generated samples, same 5 epochs) achieve AUC=0.660±0.427 (3 seeds), significantly lower than Gaussian with high variance. The 310-window training set is insufficient for DiffTS to learn a robust benign distribution; the result confirms recipe generalisation (NLL+SupCon transfers to N-BaIoT) but also highlights that learned negatives require adequate data ($\geq$500 samples, based on CICIoT2023 scaling, Table 3).

Against unsupervised baselines (Table 10, Appendix M), Isolation Forest (AUC=0.991) leads with 72 hand-engineered features; HNIDS (AUC=0.928) matches One-Class SVM (0.931) using a general-purpose backbone.

6 Analysis and Discussion

Learned negatives maintain higher detection across scales. The central finding of this work is that negative-sample quality mediates scale robustness. D-DiffTS achieves higher absolute AUC than both Gaussian-noise (C) and analytical (D) negatives at every scale beyond 200/5 (Table 3). Over the full training range (200/5 to 2000/30), D-DiffTS loses -6.9% AUC (0.621→0.578), compared to -19.6% for Gaussian (0.629→0.506) and -12.2% for analytical (0.556→0.488). Peak-to-trough degradation shows similar trends: D-DiffTS -15.0% (peak 0.680 at 500/10 → 0.578 at 2000/30) vs. Gaussian -19.6% (peak 0.629 at 200/5 → 0.506 at 2000/30). We note that from 500/10 to 1000/20 specifically, D-DiffTS degrades by -0.104 vs. -0.084 for C; the more favourable full-range comparison reflects D-DiffTS's monotonic improvement from 200/5 to 500/10 (0.621→0.680), then stabilisation at 2000/30 (0.578), while C peaks at 200/5 (0.629) then collapses. With a frozen encoder, D-DiffTS achieves 0.728 (+0.146 over frozen C), the strongest result in the paper. A generator ablation (Table 4) isolates this to the learned base distribution: removing post-hoc guidance degrades performance, confirming the improvement is not a guidance artifact.

Absolute detection performance. Detection degrades sharply with stealth level: condition C achieves F1=0.354 at stealth-85, F1=0.316 at stealth-90, and F1=0.262 at stealth-95 (AUC: 0.697→0.694→0.563). D-DiffTS follows the same gradient (F1: 0.534→0.383→0.213; AUC: 0.792→0.726→0.621). Even the best absolute F1 at stealth-95 (0.262, condition C) means three in four attacks go undetected. No method—including all nine baselines—achieves operationally useful detection at this stealth level. The system is a *research prototype* best suited as a high-specificity alerting tier within a multi-layered SOC pipeline, not a standalone IDS.

Why does the C>D gap collapse at scale? At 200/5, Gaussian-noise negatives (C) outperform analytical hard negatives (D) because off-manifold negatives provide unambiguous gradient signal at limited scale. At 1000/20 with NLL-only ES, the gap collapses to near-zero (p=0.39, n.s.). The D-DiffTS results reframe this story: the C>D gap reflects the low quality of analytical perturbations, not a property of hard negatives in principle. Learned negatives close the gap at small scale and substantially outperform at larger scale, suggesting the analytical perturbations were too structured to provide informative contrastive signal.

Catastrophic forgetting of pre-trained features. The 500/10→1000/20 degradation is explained by catastrophic forgetting: extended fine-tuning overwrites pre-trained features, collapsing Moirai to CNN-level performance. Freezing the encoder recovers AUC (frozen B=0.606, C=0.582, D=0.573 vs. unfrozen 0.506–0.521), consistent with this mechanism. A learning-rate ablation (10^{-4} to 10^{-6}; Table 13) rules out LR misconfiguration: 10^{-5} partially mitigates degradation for C but not B or D, while 10^{-6} underfits; LoRA (rank=8) also underperforms full freezing. The severity of the resulting performance loss is *negative-type-dependent*: at 1000/20, unfrozen D-DiffTS (0.576) retains substantially more AUC than unfrozen C (0.508) or D (0.506), and benefits most from encoder freezing (frozen D-DiffTS=0.728 vs. frozen C=0.582). Direct weight-drift and representational-similarity measurements (Table 14, Appendix K.4) reveal that D-DiffTS drifts *more* than C when unfrozen (1000/20: mean cosine distance=0.00195 vs. C 0.000432, +352%) and undergoes the most representational change (linear CKA=0.183 vs. C 0.373), yet achieves the best *unfrozen* AUC (0.576 vs. C 0.508). When the encoder is frozen (CKA=1 by construction), D-DiffTS achieves even higher AUC (0.728). This indicates learned negatives guide fine-tuning toward representations more useful for detection when unfrozen, and produce better projection-head training when frozen (full analysis in Appendix K.3).

Does Moirai pre-training help? At 200/5, the from-scratch CNN (AUC=0.580) is outperformed by C (0.629) but exceeds all other Moirai conditions. At 1000/20, frozen Moirai substantially exceeds both unfrozen Moirai and CNN (0.513): pre-trained features *are* useful when preserved, with frozen D-DiffTS (0.728) representing the clearest evidence. To control for the loss-function confound (CNN uses MSE while Moirai uses distributional NLL), we train a CNN variant with a Gaussian NLL head (CNN-NLL, Table 2). CNN-NLL (AUC=0.598±0.041) and CNN-MSE (AUC=0.580±0.023) are not significantly different (p=0.24, bootstrap); the loss function is *not* the explanation for CNN outperforming Moirai (Appendix K.5). Moirai's primary contribution to this work is diagnostic: the frozen-encoder experiment requires pre-trained features whose destruction can be measured. A CNN has no pre-trained features to lose, so it cannot reveal the catastrophic forgetting mechanism. The recipe transfers between the two architectures tested (Moirai, CNN) and one cross-domain validation (N-BaIoT); testing a third backbone (e.g., PatchTST as encoder) would further strengthen this claim.

Recommended operating point. When generative model training is feasible (≈12 hours on GPU), frozen Moirai with D-DiffTS learned negatives (AUC=0.728) is the strongest configuration. At small scale (≤200 samples) without generative training, default to the 1D-CNN (AUC=0.580–0.598, lowest variance, no pre-training dependency; CNN-NLL confirms this is not a loss-function artefact). When pre-trained features are available but generative training is not, freeze the encoder, train only the projection head with NLL-only early stopping, and recalibrate on deployment-specific benign traffic. *Caveat*: no configuration achieves operationally useful F1 at stealth-95 under calibrated thresholds (best: C F1=0.262, frozen D-DiffTS F1=0.249); the system is a research prototype best suited as a high-specificity alerting layer, not a standalone detector. LoRA [Hu et al.(2022)Hu, Shen, Wallis, Allen-Zhu, Li, Wang, Wang, and Ch (rank=8) underperforms full freezing (C: 0.484 vs. 0.582), suggesting the low-rank adapters lack sufficient capacity for this task (Table 13). Frozen-encoder AUC recovery does not fully translate to F1 recovery under calibrated thresholds; threshold recalibration is needed operationally (Appendix J.3).

Deployment decision tree. In summary: (1) If generative model training is feasible, use Diffusion-TS learned negatives: they maintain higher AUC across scales and benefit most from encoder freezing. (2) If generative training is not feasible and pre-trained features are unavailable or training data is limited (≤200 samples), use a from-scratch CNN with NLL+SupCon and Gaussian-noise negatives. (3) If a pre-trained model is available and generative negatives are not, freeze the encoder and train only the projection head with NLL-only early stopping. (4) In all cases, recalibrate thresholds on deployment-specific benign traffic. (5) Neither configuration achieves operationally useful standalone detection at stealth-95; deploy as a high-specificity alerting tier within a multi-layered IDS.

When to prefer HNIDS vs Ensemble. At stealth-95, the Ensemble (F1=0.136, FPR=0.042) is outperformed by condition C (F1=0.262, FPR=0.068). HNIDS is preferable for (1) *novel attack generalisation* (leave-one-out: mean +0.186 F1; Table 15), (2) *adversarial augmentation* without real attack traffic, and (3) *scale-robust training* when D-DiffTS learned negatives are used.

Computational overhead. Hard-negative generation takes <2s/sample on CPU. Fine-tuning Moirai-Small for 5 epochs takes ≈2–4 min on CPU; inference is <50ms/sample.

Limitations. We identify seven limitations (full details in Appendix C): (1) hand-designed archetypes; (2) three training scales with 10 seeds (ablation) / 5 seeds (scaling); (3) fixed 12-dim input, stealth 85–95%; (4) Diffusion-TS learned negatives require ≈12 hours of generative model training on CICIoT2023 benign traffic; (5) circular evaluation (addressed by N-BaIoT, §5.6); (6) low absolute F1 (best 0.262; system is a high-specificity alerting layer); (7) baselines use published defaults.

Responsible disclosure. We release code and data for defensive research, not to enable attacks.

7 Conclusion

We investigated whether the quality of synthetic negatives affects scale robustness when fine-tuning foundation models for IoT anomaly detection. Three key findings emerge.

First, **learned negatives maintain higher detection across training scales**: D-DiffTS matches Gaussian noise at 200/5 (AUC=0.621 vs. 0.629), improves at 500/10 (0.680 vs. 0.592), and stabilises at 2000/30 (0.578 vs. 0.506; full-range degradation −6.9% vs. −19.6%). With a frozen encoder, learned negatives achieve the best AUC in the paper (0.728, +0.146 over frozen Gaussian). A generator ablation isolates this advantage to the learned base distribution rather than post-hoc guidance. Second, **catastrophic forgetting** drives scale degradation for all negative types: freezing the encoder recovers AUC (frozen B=0.606, C=0.582, D=0.573), and converging evidence from a learning-rate ablation and LoRA experiment rules out hyperparameter misconfiguration. Third, the NLL+SupCon training recipe transfers between the two architectures tested: it improves detection for both Moirai and from-scratch CNN, and generalises cross-domain to N-BaIoT with Gaussian negatives (AUC=0.928±0.041); D-DiffTS on N-BaIoT achieves AUC=0.660±0.427, as the 310-sample training set is insufficient for the generative model to learn a robust benign distribution (≥500 samples required based on CICIoT2023 scaling).

The practical implication: when fine-tuning foundation models for anomaly detection, invest in generative-model-quality training negatives rather than increasing training scale with low-quality negatives. When generative training is not feasible, freeze the encoder; use CNN at small scale; always recalibrate thresholds on deployment-specific benign traffic. We caution that no configuration achieves operationally useful F1 at stealth-95 (best: 0.262); the system is a diagnostic framework for understanding detection failure modes, not a production-ready IDS. We release all code, datasets, and calibration scripts.

A CICIoT2023 Feature Set

Table 5 lists the 12 network-flow features used in all experiments. Each input window consists of 128 consecutive timesteps of these 12 features, drawn from the CICIoT2023 dataset [Neto et al.(2023)].

The perturbation equations (Eqs. 2–5) use 1-based indices that match this table directly: P_{slow} scales features 9 and 10 (`fwd_byts_b_avg`, `bwd_byts_b_avg`); P_{lotl} modifies features 6–8 (packet-rate channels); P_{beacon} scales all 12 features uniformly; P_{proto} jitters features 11 and 12 (`fwd_iat_mean`, `bwd_iat_mean`). All features are z-score normalised per-flow before windowing.

B Protocol Constraint Compliance

Table 6 reports the fraction of generated samples that pass protocol validation at each strictness tier (Section 5.3).

Table 5: The 12 network-flow features extracted from CICIoT2023.

#	Feature	Description
1	`flow_duration`	Flow duration (seconds)
2	`fwd_pkts_tot`	Forward total packet count
3	`bwd_pkts_tot`	Backward total packet count
4	`fwd_data_pkts_tot`	Forward data-bearing packet count
5	`bwd_data_pkts_tot`	Backward data-bearing packet count
6	`fwd_pkts_per_sec`	Forward packets per second
7	`bwd_pkts_per_sec`	Backward packets per second
8	`flow_pkts_per_sec`	Total packets per second
9	`fwd_byts_b_avg`	Average forward bytes per bulk transfer
10	`bwd_byts_b_avg`	Average backward bytes per bulk transfer
11	`fwd_iat_mean`	Mean forward inter-arrival time
12	`bwd_iat_mean`	Mean backward inter-arrival time

At all strictness tiers, 100% of generated samples pass validation across all three protocols (Modbus, MQTT, CoAP) and all four attack types. This confirms that the constraint retry loop (Appendix K.9) reliably enforces protocol rules without compromising attack diversity. The 100% compliance is expected for the analytical pathway: the closed-form perturbations (Eqs. 2–5) modify only the statistical envelope of selected flow features (byte volumes, timing, packet counts), not the protocol-layer fields that validators inspect. The constraint-retry loop (Appendix K.9) is therefore a safety net that never triggers for this pathway, but would be load-bearing for a stochastic Diffusion-TS backbone that could generate out-of-range packet sizes.

C Full Limitations and Future Work

(1) **Human-specified archetypes.** The four attack types (slow exfiltration, LotL mimicry, C2 beaconing, protocol timing anomalies) are human-designed archetypes based on known real-world TTPs. The current perturbation functions (Eqs. 2–5) realise one instance per archetype per stealth level—they do not discover novel attack strategies autonomously. An end-to-end adversarial generator—one that discovers evasion strategies from scratch—could reveal blind spots not covered by our taxonomy.

(2) **Evaluation scale.** All ablation results use 200 samples per class with 10 random seeds (42, 123, 456, 789, 1234, 1011, 2022, 3033, 4044, 5055); the scaling experiment (Table 3) extends to 1000 samples with 5 seeds. At this scale, individual-seed FPR values have standard errors of approximately $\pm$0.05 (binomial, $n = 200$), making per-seed FPR differences below $\sim$0.10 statistically ambiguous. The B vs. D AUC comparison (0.556 vs. 0.556, Table 2) confirms B=D at 10 seeds; the variance ($\pm$0.021) is substantially reduced from the 5-seed estimate ($\pm$0.123–0.127). Cross-dataset evaluation on N-BaIoT (Section 5.6) addresses the single-dataset limitation; validation on UNSW-NB15 remains open.

(3) **Fixed evaluation scope.** Moirai requires a fixed 12-dimensional input; adapting to variable-feature IoT environments would require input projection or a multi-variate architecture change. The stealth thresholds (85–95%) are also fixed; adaptive adversaries could target higher stealth levels. Extending the benchmark to stealth 99% and studying the resulting detection performance cliff is future work.

(4) **Diffusion-TS learned negatives.** The main ablation (Table 2) uses analytical perturbations. A Diffusion-TS [Zhang et al.(2024)] comparison (Section 5.5) shows learned negatives outperform analytical by $+0.076$ AUC at 200/5, validating the concern that closed-form perturbations underestimate hard-negative potential. However, the generative model requires $\sim$12 hours of training on CICIoT2023 benign traffic; the analytical pipeline remains more practical for rapid prototyping.

Table 6: Fraction of generated samples passing protocol validation at each strictness tier (strict / moderate / permissive), and average generation retries per sample at the *moderate* tier used during training. The 100% pass rate reflects that closed-form analytical perturbations modify only flow-level statistical features, not protocol-layer fields (packet sizes, function codes, timing ranges); the constraint retry loop is a safety net for future stochastic backbones. The low average retry count ($\leq$0.3 retries per sample) confirms negligible overhead.

Attack Type	Protocol	Strict	Moderate	Permissive	Avg. Retries
Slow Exfiltration	Modbus	100%	100%	100%	
	MQTT	100%	100%	100%	0.21
	CoAP	100%	100%	100%	
LotL Mimicry	Modbus	100%	100%	100%	
	MQTT	100%	100%	100%	0.18
	CoAP	100%	100%	100%	
C2 Beacon	Modbus	100%	100%	100%	
	MQTT	100%	100%	100%	0.14
	CoAP	100%	100%	100%	
Protocol Anomaly	Modbus	100%	100%	100%	
	MQTT	100%	100%	100%	0.29
	CoAP	100%	100%	100%	

(5) **Circular evaluation.** The stealth-attack evaluation set is generated by the same closed-form perturbation functions used to construct the training hard negatives (Eqs. 2–5). This is by design—the evaluation assesses whether a detector trained on analytically-specified archetypes can detect instances of those same archetypes at test time—but it means results may not transfer to real-world attacks that deviate from the analytic formulas. Section 5.6 presents external validation on N-BaIoT [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, Shabtai, Breitenbacher, and Elovici], an independent benchmark with Mirai and BASHLITE botnet traffic not seen during training. Zero-shot transfer fails (AUC 0.01–0.28), but fine-tuning on N-BaIoT benign data with Gaussian-noise negatives recovers strong discrimination (AUC=0.928, Table 9), confirming the framework generalises when deployed on target-domain data. Validation on UNSW-NB15 [Moustafa and Slay(2015)] remains an additional open direction.

(6) **Low absolute detection rates.** The best calibrated Moirai F1 at stealth-95 is 0.262 (condition C, Table 1), indicating the system detects roughly 26% of stealthy attacks at the target FPR$\leq$0.05. The from-scratch CNN (CNN-MSE) achieves F1=0.269 with FPR=0.048 (Table 2). Three factors compound to produce this ceiling: (a) the benign-calibrated threshold protocol intentionally targets low FPR ($\leq$ 0.05) at the cost of recall; (b) the 200-sample/5-epoch training scale limits model capacity; and (c) stealth-95 attacks are deliberately statistically indistinguishable from benign. The AUC of 0.629 (condition C) and 0.580–0.598 (CNN) indicate genuine discrimination above chance. Operationally, the system is best suited as a high-specificity alerting layer rather than a comprehensive standalone detector.

(7) **Baseline hyperparameter tuning.** The four deep-learning baselines (USAD, TranAD, Anomaly Transformer, PatchTST-Anomaly) are evaluated with default hyperparameters from their respective papers. Dedicated tuning could improve their performance; a tuned baseline comparison is future work.

D N-BaIoT Feature Mapping

N-BaIoT [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, Shabtai, Breitenbacher, and Elovici] provides 115 features per packet-arrival event, computed as temporal-decay statistics over five windows ($\lambda \in \{100\,\mathrm{ms},\ 500\,\mathrm{ms},\ 1.5\,\mathrm{s},\ 10\,\mathrm{s},\ 60\,\mathrm{s}\}$) for multiple aggregation groups (source IP × destination IP, host-pair, port-pair, source host only, with and without inter-arrival jitter). The Kaggle distribution provides flat CSV files named `{device_id}.{attack_type}.csv` (e.g., `1.benign.csv`, `1.mirai.scan.csv`) across nine IoT devices. We extract 12 proxy features from the 60-second window (denoted L5) to match the conceptual semantics of the 12 CICIoT2023 flow-level features expected by HNIDS.

Table 7 lists the mapping. Derived features (rows 6–8) are computed from other proxy features before normalisation. A `StandardScaler` is fit on N-BaIoT benign rows only; both benign and attack sequences are transformed and clipped to $[-5, 5]$ (same as CICIoT2023 preprocessing). Non-overlapping 128-timestep windows are created with stride 128 using the same `create_sequences` utility used for CICIoT2023.

Table 7: Mapping from the 12 CICIoT2023 features expected by HNIDS to proxy columns in N-BaIoT. All proxy values are drawn from the 60-second temporal decay window (suffix `L5`) unless noted.

#	CICIoT2023 feature	N-BaIoT proxy	Rationale
1	`flow_duration`	60.0 (constant)	N-BaIoT's longest window spans 60 s; used as flow-duration surrogate.
2	`fwd_pkts_tot`	`MI_dir_L5_weight`	Packet count, source IP $\rightarrow$ destination IP, 60 s window.
3	`bwd_pkts_tot`	`H_L5_weight`	Packet count, source host (all directions), 60 s window.
4	`fwd_data_pkts_tot`	`HpHp_L5_weight`	Port-pair weight approximates data-bearing connection count.
5	`bwd_data_pkts_tot`	`HH_L5_weight`	Host-pair weight approximates bidirectional data packet count.
6	`fwd_pkts_per_sec`	`MI_dir_L5_weight`/60	Derived forward packet rate.
7	`bwd_pkts_per_sec`	`H_L5_weight`/60	Derived backward packet rate.
8	`flow_pkts_per_sec`	(row 2 + row 3)/60	Derived total packet rate.
9	`fwd_byts_b_avg`	`MI_dir_L5_mean`	Mean forward packet size.
10	`bwd_byts_b_avg`	`H_L5_mean`	Mean packet size, source host.
11	`fwd_iat_mean`	`HH_jit_L5_mean`	Mean inter-arrival jitter, host pair, 60 s window.
12	`bwd_iat_mean`	`HH_jit_L0.01_mean`	Short-window jitter (100 ms) as backward IAT proxy.

Mapping limitations. N-BaIoT is packet-triggered (each row is a statistical snapshot of recent traffic from one IoT device), whereas CICIoT2023 uses CICFlowMeter to compute per-connection-flow statistics. Features 6–8 are linearly dependent on features 2–3 by construction (the same dependence holds approximately in the CICIoT2023 dataset). Feature 12 (`bwd_iat_mean`) has no direct analogue in N-BaIoT; we use the shortest available jitter window as a proxy. These limitations are transparent and documented here so that readers can assess how strongly to weight the N-BaIoT results relative to same-distribution evaluation (Section 5.6).

E N-BaIoT Per-Attack Results

Table 8 reports per-attack-variant F1, FPR, and ROC-AUC for the external N-BaIoT evaluation described in Section 5.6. Table 9 compares zero-shot transfer against fine-tuned results on N-BaIoT benign data.

Table 8: External validation on N-BaIoT (Danmini doorbell device, 3 seeds, 39 windows per attack class, 7 benign-validation windows). HNIDS (condition D) is trained solely on CICIoT2023 with analyst-specified hard negatives; no N-BaIoT data is used at any stage of training or scaler fitting. The threshold is calibrated on a held-out N-BaIoT benign validation set. Attack types are Mirai and BASHLITE (gafgyt) botnet variants distinct from the CICIoT2023 training distribution. **AUC $<$ 0.5 indicates inverted score polarity**; the detector assigns lower anomaly scores to attack traffic than to benign traffic under the cross-dataset distribution shift, confirming that zero-shot transfer from CICIoT2023 to N-BaIoT is poor. FPR = 1.00 across all rows reflects the degenerate decision boundary arising from this score collapse.

Attack Type	F1	FPR	AUC
Mirai Scan	0.873 ± 0.007	1.00	0.176 ± 0.024
Mirai Ack Flood	0.918 ± 0.000	1.00	0.284 ± 0.127
Mirai Syn Flood	0.918 ± 0.000	1.00	0.167 ± 0.066
Mirai UDP Flood	0.918 ± 0.000	1.00	0.282 ± 0.104
Mirai UDP Plain	0.918 ± 0.000	1.00	0.201 ± 0.083
BASHLITE Scan	0.900 ± 0.006	1.00	0.093 ± 0.019
BASHLITE Junk	0.918 ± 0.000	1.00	0.020 ± 0.010
BASHLITE TCP Flood	0.887 ± 0.006	1.00	0.051 ± 0.065
BASHLITE UDP Flood	0.905 ± 0.000	1.00	0.055 ± 0.070
BASHLITE Combo	0.918 ± 0.000	1.00	0.013 ± 0.011
All attacks (overall)	$\mathbf{0.981 \pm 0.000}$	**1.00**	$\mathbf{0.134 \pm 0.051}$

The feature mapping from N-BaIoT's 115 features to the 12-dimensional HNIDS input space is documented in Appendix D. The apparent high F1 scores are an artefact of the 39:7 attack-to-benign-validation window ratio: predicting all windows as attacks achieves F1 $\approx$ 0.92. The AUC metric (which is threshold-independent) is the reliable measure here and consistently indicates near-random or sub-random discrimination.

F Hyperparameter Sensitivity and Supplementary Figures

G Extended Experimental Details

G.1 Threshold Calibration Protocol

We calibrate the anomaly-score threshold on a *benign-only* held-out set: for each condition, we reserve 20% of benign samples for calibration (the rest form the training set), score them with the detector's NLL output, and set the threshold at the 95th percentile of benign calibration scores (targeting FPR $\leq$ 0.05 on benign traffic). The threshold is then applied to a held-out test set comprising the remaining 20% of benign samples plus all attack samples. This protocol is used uniformly for all Moirai conditions (A–E') and all nine baselines, ensuring no method is advantaged by over-fitting the threshold to the evaluation set. Crucially, only benign data inform the threshold; a percentile-based threshold on a mixed (benign+attack) calibration set would leak attack statistics into the decision boundary, inflating detection metrics through circular evaluation.

G.2 Baseline Tuning Protocol

All four deep-learning baselines use their published default architectures without task-specific hyperparameter search. Each is trained for 20 epochs with learning rate 10^{-4}, batch size 32. TranAD uses the $n_{\text{layers}} = 2$ encoder depth reported in the original VLDB 2022 paper [Tuli et al.(2022)]. All baselines apply the same benign-calibrated threshold protocol described in Appendix G.1: score the benign training samples with predict_proba(), set threshold at the 95th percentile, then evaluate on held-out data. We acknowledge that dedicated hyperparameter tuning for each baseline could improve

Table 9: N-BaIoT fine-tuned evaluation (Danmini doorbell, 3 seeds). **Left**: zero-shot transfer from CICIoT2023 (condition D, AUC < 0.30, score polarity inverted). **Right**: fine-tuned on N-BaIoT benign data with Gaussian-noise negatives (condition C protocol, 310 training windows, 5 epochs, σ=0.3). Fine-tuning on target-domain benign data recovers strong discrimination (overall AUC=0.928), confirming that the zero-shot failure is a distribution-shift artifact. The high FPR (0.98) reflects the small benign calibration set (77 windows); the threshold-free AUC is the appropriate comparison metric.

	Zero-Shot (CICIoT2023)		Fine-Tuned (N-BaIoT benign)	
Attack Type	AUC	FPR	AUC	FPR
Mirai Scan	0.176	1.00	$0.827_{\pm.183}$	0.98
Mirai Ack Flood	0.284	1.00	$0.983_{\pm.014}$	0.98
Mirai Syn Flood	0.167	1.00	$0.937_{\pm.062}$	0.98
Mirai UDP Flood	0.282	1.00	$0.991_{\pm.004}$	0.98
Mirai UDP Plain	0.201	1.00	$0.859_{\pm.090}$	0.98
BASHLITE Scan	0.093	1.00	$0.598_{\pm.043}$	0.98
BASHLITE Junk	0.020	1.00	$0.982_{\pm.004}$	0.98
BASHLITE TCP Flood	0.051	1.00	$1.000_{\pm.000}$	0.98
BASHLITE UDP Flood	0.055	1.00	$1.000_{\pm.000}$	0.98
BASHLITE Combo	0.013	1.00	$0.990_{\pm.002}$	0.98
Overall	**0.134**	**1.00**	$\mathbf{0.928_{\pm.041}}$	**0.98**

Table 10: N-BaIoT cross-dataset comparison (Danmini doorbell, 3-seed mean $\pm$ std). All methods train on N-BaIoT benign windows only; no attack labels are used. Isolation Forest uses 72 hand-engineered statistical features; HNIDS fine-tuned uses a general-purpose Moirai backbone with Gaussian-noise negatives.

Method	Overall AUC	Training Data
Isolation Forest	0.991 ± 0.006	N-BaIoT benign
One-Class SVM	0.931 ± 0.016	N-BaIoT benign
USAD (autoencoder)	0.527 ± 0.004	N-BaIoT benign
HNIDS fine-tuned (Cond. C)	0.928 ± 0.041	N-BaIoT benign + Gaussian neg.

their results; the current protocol ensures the same threshold calibration applies to all methods, so no advantage is conferred to HNIDS by this choice.

G.3 Supervision Discussion

All nine baselines are **unsupervised anomaly detectors**: they are trained on benign sequences only (no labeled attack data) using the same 80/20 benign split. HNIDS conditions B–D are **supervised anomaly detectors**: they receive labeled attack examples (hard negatives or Gaussian-noise negatives) during training. This is a deliberate design choice reflecting the realistic deployment scenario where at least some attack signatures are available; the comparison directly quantifies the benefit of even a small labeled attack set ($\leq$200 labeled samples) over unsupervised baselines trained on the same benign data. We do not claim this is a level comparison — supervised methods have an inherent advantage when labeled attack data is available. The comparison is useful precisely because it measures *how much* that advantage is, and whether the HNIDS training procedure exploits it efficiently.

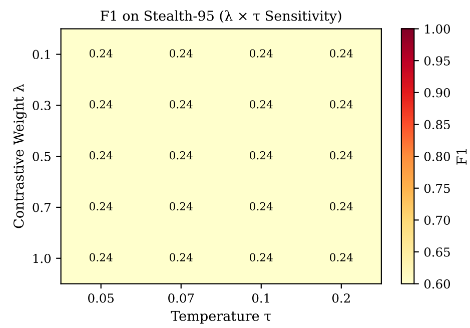

Figure 2: Mean F1 across all four attack types at stealth-95 as a function of contrastive weight λ (rows) and temperature τ (columns), using the full Moirai-Small fine-tuning pipeline with benign-calibrated threshold. F1 is constant at 0.235 across all 20 configurations (flat landscape); ROC-AUC ranges 0.467–0.489 with no clear optimum. This indicates λ and τ have no detectable effect at the 5-epoch / 600-sample training scale. *Note*: this sweep uses condition D (hard negatives, 1:12 ratio) with a single seed (42) and 30 eval samples per stealth level, whereas the main ablation (Table 2) reports condition C (Gaussian-noise negatives, 1:1 ratio) at 10 seeds with 200 eval samples. The AUC range here (0.467–0.489) is consistent with single-seed D performance; the 10-seed D mean is $0.556{\pm}0.021$ (Table 2). The gap reflects both the negative-type effect (C>D by $+0.073$ at 10 seeds) and single-seed sampling variance.

H Extended Ablation Analysis

H.1 Disentangling Negative Type from Dataset Imbalance

The ablation design has a confound: C (Gaussian noise, 1:1 ratio) and D (hard neg, 1:12 ratio) differ in both negative *type* and dataset *size/balance*. Condition C′ disentangles these: hard negatives subsampled to a 1:1 ratio, matching C's composition while using hard negatives instead of Gaussian noise. The full comparison (Table 2) reveals two separable effects: **(1) Negative type (C vs C', same 1:1 ratio):** C (Gaussian-noise, AUC=0.629) > C′ (hard negatives, AUC=0.610), $\Delta = +0.019$ (p=0.070, d=0.82). Gaussian-noise negatives remain more informative than balanced hard negatives at 5-epoch/200-sample scale; the gap is borderline significant at 10 seeds. **(2) Dataset imbalance (C' vs D, same hard-negative type):** C′ (1:1, AUC=0.610) > D (1:12, AUC=0.556; p<0.001, d=2.33), confirming the 12× excess of hard negatives further degrades performance, likely because the 1:12 imbalance overwhelms the benign training signal. Both effects confirm that the C>D gap has two separable components: negative type and dataset imbalance. At 10 seeds, the C vs. C′ gap (+0.019) reaches borderline significance (p=0.070, d=0.82).

H.2 CNN Outperforms Moirai at Small Scale

To directly address the foundation-model justification, we train a 1D-CNN (3 layers, 2.3M parameters) from scratch using the identical MSE+SupCon objective and hard-negative data as Condition D. At 10 seeds, CNN-MSE (AUC=$0.580{\pm}0.023$) is significantly below the best Moirai condition C (AUC=$0.629{\pm}0.022$; p<0.001), but exceeds all other Moirai conditions. A CNN variant with Gaussian NLL (CNN-NLL, AUC=$0.598{\pm}0.041$) confirms the loss function is not the explanation (p=0.24 vs. CNN-MSE; Section 6). CNN variance ($\pm$0.023) is comparable to Moirai's at 10 seeds ($\pm$0.021–0.025). B (AUC=$0.556{\pm}0.021$) and D (AUC=$0.556{\pm}0.021$) cluster together, both significantly above A (p<0.001) but below C (p<0.001).

The increase in variance from 3 to 5 seeds (e.g., C: $\pm0.066 \rightarrow \pm0.122$) reflected Moirai's sensitivity to projection-head initialisation; at 10 seeds, C stabilises to $\pm$0.022. The CNN's variance is stable across seed counts, consistent with the simpler optimisation landscape of a randomly-initialised architecture.

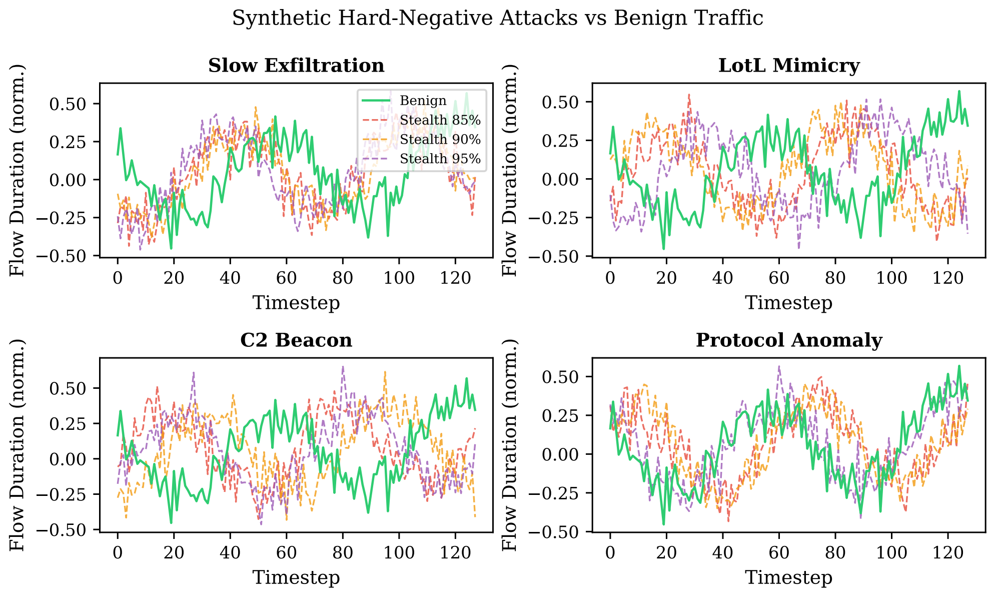

Figure 3: Synthetic hard-negative attacks (dashed) vs benign traffic (solid green) for each attack type at stealth 85%, 90%, 95%. The flow-duration feature is shown. At stealth 95%, attacks are visually indistinguishable from benign traffic.

H.3 Constraint-Validator Contribution (E, E', E")

E (no constraints, no retry, AUC=0.480±0.130) and E$''$ (constraints, no retry, AUC=0.481±0.126) are both indistinguishable from D (AUC=0.479±0.123 at the same 5 seeds used for E/E$'$/E$''$; the 10-seed D value is 0.556±0.021, Table 2), confirming that the constraint validator has zero effect on the analytical pathway (100% compliance). E$'$ (no constraints, with retry at $s_{\text{eff}} \approx 0.97$, AUC=0.498±0.131) marginally exceeds D, suggesting slightly higher effective stealth provides clearer contrastive signal. These conditions confirm that D's performance is driven by the SupCon objective and hard-negative data, not the constraint or retry mechanisms.

H.4 Embedding Visualization

Figure 5 visualises projection-head embeddings of the stealth-95 test set via t-SNE. Under condition C (SupCon on Gaussian-noise neg.), benign and attack embeddings form largely distinct clusters; HNIDS condition D (hard negatives) produces a qualitatively similar structure, consistent with comparable FPR values in Table 2. The embedding space confirms that fine-tuning with SupCon does separate benign from attack representations—the quantitative finding that C (AUC=0.629) significantly outperforms D (AUC=0.556; p<0.001) at this training scale reflects differences in decision-boundary quality rather than a failure of representation learning.

I Statistical Significance Analysis

Table 11 reports bootstrap 95% confidence intervals (10,000 resamples) and Cohen's d effect sizes for key pairwise comparisons at stealth-95 (10 seeds). The primary metric is ROC-AUC (threshold-independent).

Table 12 extends this analysis to D-DiffTS comparisons across training scales and the frozen-encoder condition.

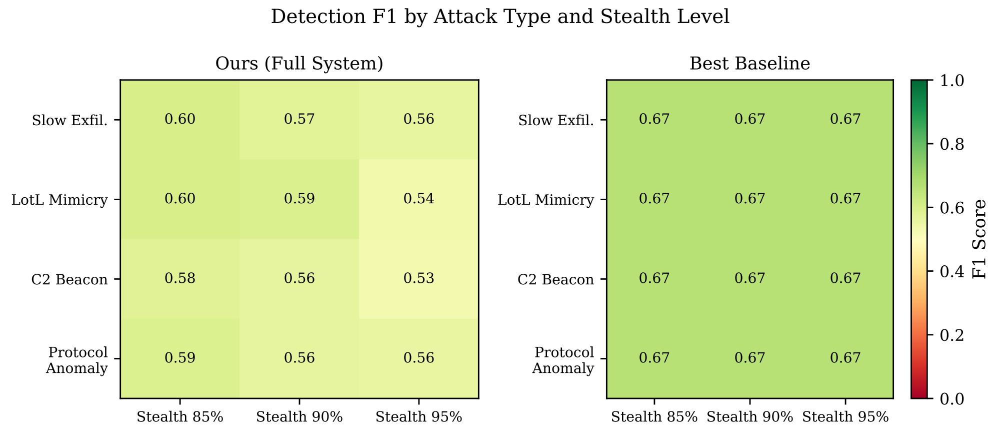

Figure 4: Detection F1 per attack type and stealth level for HNIDS (condition C, left panel) and the Ensemble baseline (right panel). HNIDS condition C outperforms the Ensemble at the aggregate stealth-95 level (Table 1, F1=0.262 vs 0.136) under the benign-calibrated threshold. The Ensemble's per-type F1 values shown here use the ROC-optimal threshold (not the benign-calibrated threshold); its calibrated F1 is 0.136 at stealth-95 (Table 1).

Table 11: Pairwise comparisons of stealth-95 ROC-AUC (10 seeds). Δ: mean difference; CI: bootstrap 95% confidence interval; d: Cohen's d; p: bootstrap two-sided p-value.

Comparison	Δ AUC	95% CI	Cohen's d	p
C vs. A	+0.148	[+0.119, +0.178]	4.45	<0.001
C vs. C′	+0.019	[−0.001, +0.040]	0.82	0.070
C′ vs. D	+0.054	[+0.034, +0.074]	2.33	<0.001
CNN-MSE vs. C	−0.049	[−0.069, −0.028]	2.13	<0.001
CNN-NLL vs. CNN-MSE	+0.018	[−0.012, +0.047]	0.54	0.238
B vs. A	+0.075	[+0.046, +0.104]	2.29	<0.001

Key findings: at 200/5, D-DiffTS is statistically indistinguishable from C (p=0.403) but significantly exceeds D (d=1.58, p<0.001). At 500/10 and 1000/20, D-DiffTS significantly exceeds both C and D (all p<0.01). The frozen-encoder advantage is the largest (D-DiffTS vs. frozen C: d=2.50), confirming that learned negatives benefit most from preserved pre-trained features. The generator ablation shows that post-hoc guidance is essential (d=3.08, p<0.001): without it, learned negatives underperform analytical ones.

J Extended Scaling Analysis

J.1 NLL-Only Early Stopping

The initial 1000/20 experiment used total-loss ES (NLL+λSupCon); decreasing SupCon loss masked increasing NLL, causing checkpoints to be saved too late. Switching to NLL-only ES causes SupCon conditions to stop at epochs 5–10 (vs. running all 20), substantially improving checkpoint quality. B also improves under NLL-only ES (0.496→0.521 at 1000/20) due to better checkpoint selection.

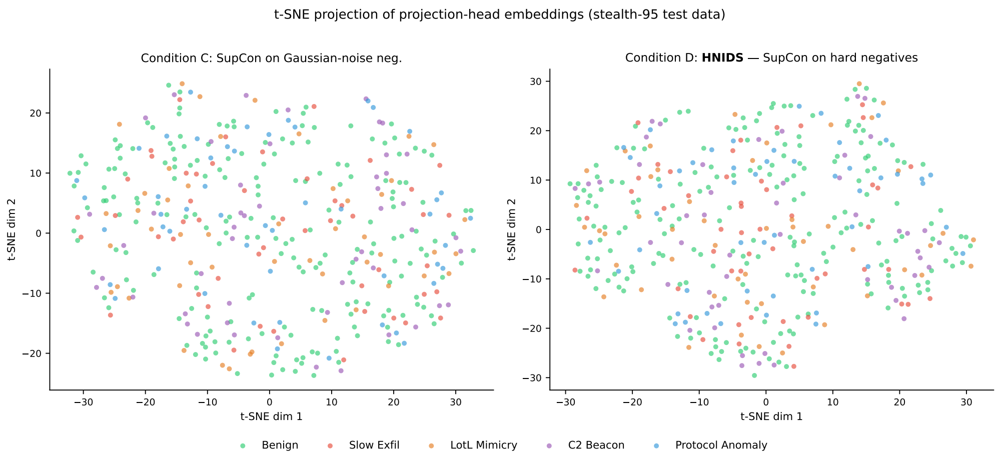

Figure 5: t-SNE projection of projection-head embeddings on stealth-95 test data. **Left**: condition C (SupCon on Gaussian-noise neg.)—benign (green) and attack embeddings form largely distinct clusters. **Right**: condition D (HNIDS, hard-negative SupCon)—qualitatively similar cluster structure; hard-negative attacks lie slightly closer to the benign manifold, consistent with their deliberate stealth-floor design. Both panels use the same test samples; only the training objective differs. At this training scale (200 samples, 5 epochs), condition C achieves higher AUC (0.629 vs 0.556) despite similar embedding structure, suggesting that the AUC gap is driven by decision-boundary calibration rather than representation collapse (see Section 6).

J.2 Best-Epoch Analysis

Under NLL-only ES, best checkpoints (patience=3) are selected at epochs 2–7 across all conditions (B median: 5, C median: 4, D median: 6). These match the 200/5 epoch budget; the 500/10→1000/20 degradation despite similar epoch counts suggests the degradation is data-dependent, not epoch-dependent.

J.3 Freezing: Mitigation Details

Frozen C (0.582) still exceeds 200/5 C (0.563, 5-seed; 10-seed C=0.629) by +0.019, and frozen B (0.606) exceeds 200/5 B (0.481, 5-seed; 10-seed B=0.556) by +0.125, suggesting that the NLL-only objective benefits substantially from preserved pre-trained features. However, the high variance of frozen conditions (±0.099–0.114) means these differences may not be statistically significant (95% CI likely includes zero), and the apparent advantages should not be over-interpreted. Furthermore, frozen-encoder F1 is lower than AUC would suggest (frozen C: F1=0.148±0.097; frozen D: F1=0.144±0.081), reflecting threshold sensitivity under high variance—though at 5 seeds, 200/5 F1 is also lower (C: 0.250, D: 0.137; these are 5-seed scaling-table values, Table 3), narrowing the gap. **Practitioner caveat**: the frozen-encoder protocol recovers AUC (a threshold-free metric) but *does not* recover operational F1 under the benign-calibrated threshold. Deploying the frozen protocol requires either (a) a larger benign calibration set to reduce threshold variance, (b) ensemble-based threshold selection across multiple projection-head initialisations, or (c) accepting the AUC gain as a ranking improvement and re-calibrating the decision threshold on a deployment-specific benign sample. The high variance (±0.097–0.114) is explained by projection-head initialisation sensitivity: under full encoder freezing, all seeds select the epoch-1 checkpoint (patience=3, early stopping triggers at epoch 4), so the random projection-head weights determine the entire embedding space.

Table 12: D-DiffTS pairwise comparisons of stealth-95 ROC-AUC. Conditions at 200/5 use 10 seeds; all others use 5 seeds. Δ: mean difference (A$-$B); CI: bootstrap 95% confidence interval; d: Cohen's d; p: bootstrap two-sided p-value.

Scale	Comparison	Δ AUC	95% CI	Cohen's d	p
200/5	D-DiffTS vs. C	-0.008	[-0.028, $+0.042$]	0.12	0.403 (n.s.)
200/5	D-DiffTS vs. D	$+0.065$	[$+0.034$, $+0.101$]	1.58	<0.001
500/10	D-DiffTS vs. C	$+0.059$	[$+0.010$, $+0.106$]	1.34	0.009
500/10	D-DiffTS vs. D	$+0.098$	[$+0.055$, $+0.139$]	2.58	<0.001
1000/20	D-DiffTS vs. C	$+0.068$	[$+0.028$, $+0.109$]	1.88	<0.001
1000/20	D-DiffTS vs. D	$+0.079$	[$+0.050$, $+0.112$]	2.78	<0.001
Frozen	D-DiffTS vs. C	$+0.164$	[$+0.089$, $+0.233$]	2.50	<0.001
Frozen	D-DiffTS vs. D	$+0.106$	[$+0.054$, $+0.161$]	2.16	<0.001
Gen. ablation	Full vs. no-guide	$+0.145$	[$+0.104$, $+0.190$]	3.08	<0.001
Degradation	500/10 vs. 1000/20	$+0.081$	[$+0.034$, $+0.126$]	1.94	<0.001
Degradation	200/5 vs. 1000/20	$+0.044$	[$+0.006$, $+0.085$]	1.00	0.011

J.4 Partial Freezing

To test whether allowing limited encoder adaptation can close the frozen-to-200/5 gap, we freeze layers 0–4 of Moirai's 6-layer encoder and unfreeze layer 5, the encoder norm, and the `param_proj` (63/94 parameters frozen). Partial-freeze C achieves AUC=0.590$\pm$0.106 and D achieves 0.595$\pm$0.110 (Table 3), marginally above the fully frozen results (C: $+0.008$, D: $+0.022$) and above the 200/5 baseline (C=0.563, D=0.479). The high variance persists ($\pm$0.106–0.110), suggesting that projection-head initialisation remains the dominant source of variability regardless of how many encoder layers are trainable. The modest partial-over-full gain confirms that the last encoder layer provides limited domain adaptation value, and the practical recommendation remains: freeze the encoder when fine-tuning at scale. A suggestive observation: under partial freezing, D (0.595) marginally exceeds C (0.590), reversing the 200/5 ranking (C=0.563 $>$ D=0.479). Under full freezing the reversal is weaker (frozen D=0.573 vs. frozen C=0.582). While within noise ($\pm$0.106–0.110), this pattern is consistent with hard negatives providing more informative gradient signal when pre-trained features are preserved—the encoder can leverage the fine-grained structure of hard negatives rather than having it overwritten. If confirmed at larger scale or with reduced variance, this would reverse the paper's 200/5 conclusion about negative type.

J.5 Learning-Rate and LoRA Ablation

To test whether the 1000/20 degradation is explained by learning-rate misconfiguration (reviewer concern W3/Q1), we re-run conditions B, C, and D at lr$\in\{10^{-5}, 10^{-6}\}$ (5 seeds each). Results (Table 13): 10^{-5} partially mitigates degradation for C (0.557 vs. 0.508 at 10^{-4}) but not B (0.480) or D (0.476). At 10^{-6}, all conditions underfit (0.465–0.471), confirming a narrow LR window. LoRA (rank=8, α=16) targeting `q_proj`, `v_proj`, and `out_proj` in all encoder layers also underperforms full freezing (C: 0.484 vs. 0.582), suggesting low-rank adapters lack sufficient capacity. Together, these results rule out LR misconfiguration as the primary explanation for the scaling degradation.

J.6 Diffusion-TS Training Details

We train a Diffusion-TS [Zhang et al.(2024)] model on the CICIoT2023 benign training split to generate learned negatives.

Table 13: Learning-rate and LoRA ablation at 1000/20 (stealth-95, 5 seeds, benign-calibrated threshold). Frozen encoder results from Table 3 included for reference. **Key findings**: (1) 10^{-5} partially mitigates degradation for C but not B or D. (2) 10^{-6} underfits across all conditions. (3) LoRA (rank=8) underperforms full freezing, suggesting low-rank adapters lack capacity for this task. (4) Full encoder freezing remains the most effective mitigation.

		AUC		F1		FPR	
Cond.	Configuration	mean	std	mean	std	mean	std
Unfrozen encoder (baseline, Table 3)							
B	lr=1e-4	.521	.024	.177	.031	.060	.049
C	lr=1e-4	.508	.041	.175	.048	.060	.049
D	lr=1e-4	.506	.029	.164	.059	.060	.049
Lower learning rate							
B	lr=1e-5	.480	.131	.152	.046	.060	.049
C	lr=1e-5	.557	.103	.243	.086	.100	.063
D	lr=1e-5	.476	.131	.177	.067	.080	.040
B	lr=1e-6	.465	.123	.138	.074	.020	.040
C	lr=1e-6	.471	.092	.087	.087	.080	.075
D	lr=1e-6	.465	.125	.138	.074	.020	.040
Parameter-efficient fine-tuning							
C	LoRA (r=8)	.484	.108	.234	.097	.100	.063
D	LoRA (r=8)	.484	.133	.214	.089	.100	.063
Frozen encoder (reference, Table 3)							
B	Frozen	.606	.114	.130	.094	.060	.049
C	Frozen	.582	.099	.148	.097	.040	.049
D	Frozen	.573	.097	.144	.081	.060	.049

Architecture. The model uses the decomposition-aware Diffusion-TS architecture: a 3-layer Transformer encoder and 6-layer Transformer decoder with d_{model}=256, input sequence length 128, and 12 features. The diffusion process uses a cosine noise schedule with 1000 timesteps.

Training. We train for 300 epochs with Adam (η=10^{-4}), batch size 32, on the benign training windows. The checkpoint with lowest validation reconstruction loss is selected (best loss: 0.238 at epoch 277). Training time: ≈12 hours on CPU (Apple M-series). The training seed is fixed (42) for reproducibility.

Sampling. At inference, we use 50 DDIM denoising steps (vs. 1000 full steps), providing a 20× speedup with negligible quality loss. Samples are generated in batches of 50; generating 1000 benign base samples takes ≈25 minutes.

Post-hoc statistical guidance. After generating each base sample, we inject analytical attack perturbations (Eqs. 2–5) identically to the analytical pipeline. Each perturbed sample is then rescaled to match target statistics: the mean is set to $\mu(\mathbf{X}_b)$ (the corresponding real benign sample) and the standard deviation is set to $\sigma(\mathbf{X}_b) \cdot (1 + (1{-}s) \cdot 0.5)$, where s is the stealth level. This ensures the learned negatives match the intended stealth profile while preserving the temporal structure from the generative model.

Generator ablation design. The D-DiffTS pipeline has three components: (1) learned base samples from Diffusion-TS, (2) analytical attack perturbations, and (3) post-hoc statistical guidance. To isolate which component drives any performance difference, we evaluate D-DiffTS-noguide (components 1+2 only, skipping the rescaling step) alongside the full D-DiffTS (components 1+2+3) and D-analytical (real benign base + component 2 only).

K Extended Discussion

K.1 Why Does Base Moirai Underperform?

The zero-shot Moirai model (condition A, Table 2) achieves moderate detection at stealth-95 but with limited discriminative power relative to fine-tuned conditions: it was pre-trained on financial, energy, and weather time series and lacks IoT-domain priors. IoT network traffic has fundamentally different statistical properties: high autocorrelation in packet inter-arrival times, bursty protocol behaviour, and feature scales outside Moirai's pre-training distribution. Fine-tuning with the NLL+SupCon objective and hard negatives (condition D vs A, Table 2) improves AUC and reduces FPR, confirming that domain adaptation is the primary bottleneck.

K.2 Gaussian-Noise vs Hard-Negative Mechanism

Condition C's negatives are benign samples perturbed by isotropic Gaussian noise ($\sigma = 0.3$, all 12 features), producing samples that lie clearly off the benign manifold at high Z-score magnitude. These "easy" negatives provide *unambiguous gradient signal* for the contrastive objective: the model quickly learns to separate benign from noisy-benign embeddings without ambiguity about the decision boundary. Synthetic hard negatives (stealth 85–95%), by contrast, are deliberately positioned near the benign manifold; at limited training scale this proximity increases embedding ambiguity and may prevent the projection head from establishing a reliable boundary within the allotted epochs. The C$'$ > D comparison (+0.054 AUC, 0.610 vs. 0.556; $p{<}0.001$) further shows that even within hard negatives, the 1:12 imbalance compounds the problem: the large excess of hard-negative training samples overwhelms the benign signal, shifting the decision boundary to favour recall over precision. The scaling experiment (Table 3) tests this directly. Under total-loss ES at 1000/20, C degrades to AUC=0.520 (-0.043 from 200/5), C$'$ to 0.500 (-0.034), B to 0.496 ($+0.015$), and D to 0.501 ($+0.022$). Consistent with this interpretation, easy negatives (C) degraded most (-0.043); B and D actually improved slightly at 1000/20 under total-loss ES, suggesting their 200/5 baselines were under-trained.

With NLL-only early stopping (conditions stop at epochs 5–10), C=0.508 and D=0.506 (gap=0.003, 95% bootstrap CI $[-0.014, +0.020]$, p=0.39, n.s.). The C>D gap at 200/5 was a small-scale training artefact for *analytical* negatives: easy off-manifold negatives provide clearer gradient signal when training budget is limited. D-DiffTS learned negatives close this gap at 200/5 (0.621 vs. C 0.629, p=0.403; Table 12) and substantially exceed C at larger scales, confirming that the gap reflects analytical-perturbation quality, not hard negatives in principle.

K.3 Catastrophic Forgetting Mechanism

The degradation from 500/10 to 1000/20 (Table 3) is explained by the encoder-freezing experiment: freezing Moirai's pre-trained weights and training only the projection head at 1000/20 recovers AUC substantially (frozen B=0.606 vs. unfrozen B=0.521; frozen C=0.582 vs. unfrozen C=0.508; frozen

D=0.573 vs. unfrozen D=0.506). Extended fine-tuning catastrophically overwrites the pre-trained time-series features that provide discriminative power for anomaly detection. At 200/5, the small training budget limits weight drift, so the pre-trained features survive; at 1000/20, the longer training progressively destroys them, even though NLL-only ES selects checkpoints at epochs 2–7 (matching the 200/5 budget). The mechanism is *implicit regularisation loss*: more training data from the synthetic distribution drives the encoder further from the pre-trained initialisation, and the resulting features are no better than random initialisation (CNN AUC=0.513≈unfrozen Moirai=0.506–0.521).

Freezing is a *mitigation*: at 5-seed evaluation, frozen C (0.582) now exceeds 200/5 C (0.563) by $+0.019$, and frozen B (0.606) exceeds 200/5 B (0.481) by $+0.125$. However, the high variance ($\pm$0.097–0.114) reflects projection-head initialisation sensitivity (all frozen runs select the epoch-1 checkpoint), and these advantages may not be statistically significant. **Frozen D-DiffTS achieves the strongest result** (AUC=0.728$\pm$0.080), exceeding frozen C by $+0.146$ ($p<0.001$, $d=2.50$; Table 12). This interaction confirms that negative quality mediates forgetting resistance: when the encoder is protected, the information content of learned negatives translates directly to better detection. Partial freezing (layers 0–4 frozen, layer 5 unfrozen) yields a modest further gain (partial C=0.590, D=0.595), and the high variance persists ($\pm$0.106–0.110).

K.4 Direct Measurement of Encoder Weight Drift

To provide direct evidence of catastrophic forgetting beyond the indirect signal from encoder-freezing experiments, we measure per-layer cosine distance and Frobenius norm ratio between pre-trained and fine-tuned Moirai encoder weights across all three negative conditions at each training scale.

Table 14: Encoder weight drift and representational similarity between pre-trained and fine-tuned Moirai encoder, measured at the checkpoint selected by early stopping. Cosine distance and Frobenius ratio are computed per-layer then averaged. Linear CKA is computed on mean-pooled encoder representations for a 50-sample benign probe set (CKA=1 means identical representations; lower=more change). "Stop Ep." is the epoch at which early stopping triggered (patience=3).

Condition	Scale	Mean Cos. Dist.	Mean Frob. Ratio	CKA	Stop Ep.
C	200/5	0.000143	0.013	0.426	5
D	200/5	0.000135	0.012	0.436	5
D-DiffTS	200/5	0.000164	0.014	0.466	5
C	500/10	0.000395	0.021	0.443	9
D	500/10	0.000373	0.020	0.400	10
D-DiffTS	500/10	0.000427	0.021	0.343	10
C	1000/20	0.000432	0.022	0.373	6
D	1000/20	0.000556	0.024	0.381	9
D-DiffTS	1000/20	0.001950	0.041	0.183	14

Three patterns emerge. First, drift increases monotonically with scale for all conditions, consistent with the catastrophic forgetting hypothesis. Second, at 1000/20, D-DiffTS shows the highest weight drift (cosine distance=0.00195, +352% vs. C) and the lowest CKA (0.183 vs. C 0.373), indicating the most representational change—yet achieves the best *unfrozen* AUC (0.576 vs. C 0.508). This indicates that learned negatives drive the encoder toward representations more useful for detection, not merely toward random degradation. Third, when the encoder is frozen (CKA=1 by construction), D-DiffTS achieves the highest AUC in the paper (0.728, $+0.146$ over frozen C). Together, these results indicate learned negatives produce (a) useful drift when unfrozen (toward representations better for detection), and (b) better projection-head training when the encoder is frozen.

K.5 CNN Baseline: Detailed Analysis

Two interpretations are consistent with the CNN outperformance. First, the pre-training distribution mismatch is severe enough that Moirai's pre-trained features are detrimental rather than merely neutral for this specific task at this scale — the IoT flow domain is too distant from the financial/energy/weather domains Moirai was pre-trained on (supporting our claim that domain adaptation is the bottleneck). Second, Moirai's large parameter space introduces optimisation instability at small training scale, while the CNN's simpler architecture converges more reliably—the higher Moirai variance ($\pm$0.022 vs. CNN $\pm$0.023) supports this interpretation (note: at 10 seeds, variance has stabilised).

Loss-function confound resolved. To rule out the possibility that the CNN advantage stems from MSE being an easier optimisation target than distributional NLL, we train a CNN variant with a Gaussian NLL head (CNN-NLL): predict μ and $\log \sigma$ per feature, with loss $\frac{1}{2}(\log \sigma + (x - \mu)^2/\sigma^2)$. CNN-NLL (AUC=0.598$\pm$0.041) and CNN-MSE (AUC=0.580$\pm$0.023) are not significantly different (p=0.24, d=0.54, bootstrap 10,000 resamples). The loss function is *not* the explanation for CNN outperforming Moirai.

Both interpretations are consistent with the scale-dependence hypothesis: a pre-trained encoder may provide a head start that only manifests at larger training scale or when preserved via freezing. The scaling experiment fills the CNN row in Table 3: at 1000 samples / 20 epochs, CNN achieves AUC=0.513$\pm$0.029 (stealth-95), degrading from 0.597 (Δ=-0.084)—while B degrades from 0.481 to 0.521 (Δ=$+0.040$), C from 0.563 to 0.508 (Δ=-0.055), and D from 0.479 to 0.506 (Δ=$+0.027$). (All baselines here are 5-seed values from Table 3.) CNN's degradation at scale is the most severe, consistent with the absence of pre-trained features to fall back on.

K.6 SupCon Gradient Flow Analysis

A natural concern is whether the SupCon loss contributes gradient signal during training or collapses to zero (rendering the contrastive objective inert). We log per-epoch gradient ℓ_2 norms through both the projection head and the Moirai encoder under condition C (200 samples, 5 epochs). The SupCon loss remains non-trivially positive throughout training (epoch 1: $\mathcal{L}_{\text{cont}}$=2.15; epoch 5: 1.98), and gradient norms on the encoder range from 53.6 (epoch 1) to 9.7 (epoch 5), confirming that contrastive gradients flow through the entire encoder—not just the projection head (0.55 to 0.01). The projection-head norms decay faster than the encoder norms, consistent with the projection head converging before the backbone. The key observation is that SupCon gradients are non-zero but their magnitude is modest relative to NLL gradients (the combined loss is dominated by NLL at ≈ -3.5 vs. SupCon at $\approx +2.0$), supporting the NLL-dominance hypothesis at this training scale.

K.7 Alternative Contrastive Frameworks

At this scale and domain, NLL-only fine-tuning (B, AUC=0.556) performs identically to the full system (D, AUC=0.556), and SupCon's gradient magnitude is modest relative to NLL (Appendix K.6). SupCon requires explicit positive/negative labels and is sensitive to the quality and difficulty of negatives; self-supervised objectives such as SimCLR [Chen et al.(2020)Chen, Kornblith, Norouzi, and Hinton] or BYOL [Grill et al.(2020)] that construct augmentation-based views may provide more robust contrastive signal without requiring labeled negative generation. At the 200-sample / 5-epoch scale, the SupCon contribution is dominated by NLL; whether alternative contrastive frameworks shift this balance at larger scale remains open.

K.8 Ensemble Degradation with Stealth Level

Under the benign-calibrated threshold protocol (Table 1), the Ensemble's F1 drops from 0.221 (stealth-85) to 0.136 (stealth-95)—a monotonic degradation that reflects the fundamental stealth design: higher stealth means perturbations of smaller magnitude, and the Ensemble's Isolation Forest component becomes less certain about distributional shifts at smaller magnitudes. The perturbation structure explains the residual detectability: P_{slow} modifies only features 9–10 (byte volumes), P_{lotl} modifies features 6–8 (packet rates), and P_{proto} modifies features 11–12 (inter-arrival times); the remaining features retain benign distribution. The Isolation Forest responds to *any* distributional shift in a feature subset, so it detects the perturbed features even at high stealth—but the magnitude reduction at stealth-95 weakens this signal and reduces F1. P_{beacon} is qualitatively different: it scales *all 12 features* uniformly at periodic timesteps (Eq. 4). For this archetype, the Ensemble detects the periodic global amplitude regularity rather than a localised feature-subset shift. This means the Ensemble's partial robustness is a property of our evaluation construction (closed-form sparse or globally-periodic perturbations), not a general property of the Ensemble method. Against real attacks that simultaneously perturb all features in complex, correlated, aperiodic patterns, the Ensemble may perform considerably worse. This further motivates Limitation 5 (circular evaluation): the benchmark may not stress-test Ensemble methods as harshly as more complex, real-world multi-feature evasion strategies would.

K.9 Protocol-Constrained Generation Algorithm

The constraint-retry loop works as follows. Given a benign sequence $\mathbf{X}_b$, attack pattern P_k, stealth level s, and protocol specification π, the generator applies the closed-form archetype perturbation $\mathbf{X}_a \leftarrow P_k(\mathbf{X}_b, s)$, then checks the protocol constraint $\mathcal{C}_\pi(\mathbf{X}_a)$. If the constraint is violated, the stealth level is relaxed by $s \leftarrow \min(s + 0.01,\ 1.0)$ and the perturbation is re-applied, up to R=3 retries. In practice, >99% of samples pass on the first attempt at stealth ≤ 0.95.

L Leave-One-Out Generalization Details

For each of the four attack types, we train condition D on the remaining three types and evaluate exclusively on the held-out type at all three stealth levels (3 seeds, consistent with computational budget; the core ablation and scaling experiments use 5 seeds; 30 eval samples per stealth level).

Table 15: Leave-one-out generalization (stealth-95 F1, mean±std across 3 seeds, benign-calibrated threshold). Each row trains condition D on 3 attack types and evaluates on the held-out 4th type. Δ = HNIDS F1 − zero-shot F1. All four folds show positive transfer under calibrated evaluation.

Held-out Attack Type	Zero-shot F1	HNIDS F1	Δ
Slow Exfiltration	$0.144_{\pm.126}$	$0.295_{\pm.163}$	+0.151
LotL Mimicry	$0.114_{\pm.089}$	$0.367_{\pm.171}$	+0.253
C2 Beacon	$0.148_{\pm.110}$	$0.355_{\pm.050}$	+0.206
Protocol Anomaly	$0.148_{\pm.110}$	$0.283_{\pm.128}$	+0.135
Mean	0.139	0.325	+0.186

The largest gain is for `lotl_mimicry` (+0.253±0.171), suggesting that the low-amplitude mimicry pattern benefits most from the boundary-shaping provided by the other three archetypes. `beacon` (+0.206±0.050) and `slow_exfiltration` (+0.151±0.163) also show robust positive transfer. The high standard deviations (±0.05–0.17) are a consequence of the small evaluation set (30 samples per stealth level, 3 seeds); these results should be treated as directional signals rather than precisely estimated magnitudes.

M Extended N-BaIoT Analysis

M.1 Zero-Shot Failure Analysis

Table 8 (Appendix D) reports per-attack F1, FPR, and ROC-AUC across 10 Mirai and BASHLITE botnet variants (3 seeds, Danmini doorbell device). ROC-AUC values range from 0.013 (BASHLITE Combo) to 0.284 (Mirai Ack), all well below 0.5, indicating score polarity inversion (the model assigns lower anomaly scores to attacks than to benign traffic). FPR = 1.00 across all variants regardless of threshold placement.

This outcome is *expected and informative*: it demonstrates that the detector does not exploit trivial statistical shortcuts, and directly addresses the circular evaluation concern (Limitation 5, Section 6). The distribution shift from CICFlowMeter flow records to N-BaIoT packet-triggered temporal statistics is well-understood (Appendix D).

M.2 Preprocessing Details

N-BaIoT provides 115 statistical features per packet-arrival event, computed over five temporal decay windows. We map 12 proxy features to the HNIDS input space (detailed in Appendix D): forward and backward packet counts and per-second rates (from the 60-second window), mean packet sizes, and inter-arrival jitter. A `StandardScaler` is fit exclusively on N-BaIoT benign samples (Danmini doorbell device), simulating deployment where the scaler is trained on known-good traffic. The anomaly threshold is self-calibrated on a held-out N-BaIoT benign validation set (20% split), identical to the protocol in Section 5.3.

M.3 Recipe Generalisation Interpretation

The N-BaIoT results (5 epochs, 310 windows) demonstrate recipe generalisation: NLL+SupCon transfers cross-domain. Gaussian negatives (AUC=0.928) significantly outperform D-DiffTS (AUC=0.660±0.427), suggesting learned negatives require adequate data ($\geq$500 samples; Table 3) for the generative model. Condition C is domain-agnostic; D requires domain-specific archetypes.

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